

Separating the glider from the air: paraglider sink polars and the flow field of an Alpine valley from 86 000 flight logs

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Abstract

A variometer records the vertical motion of the air plus the sink of the glider. Flight logs therefore measure the air only if the sink polar is known; for paragliders it is not. We measure sink polars of 101 glider types from 25 341 of 86 190 flight logs recorded in the Valais (Swiss Alps) in 2007–2026. At fixed airspeed the sink is constant and the air varies, so the most frequent vertical speed in straight flight estimates the sink. The method assumes that at the airspeeds they fly most, pilots glide through ordinary air, and keep other speeds for special air: slow in lift, fast in strong sink and when leaving a thermal. We correct for density and for the subsidence of the glide air, 0.12 m s^{-1} (95 %: 0.08–0.14) per 1 m s^{-1} of daily thermal strength, and fit $s = av^3 + b/v$ to 1 km h^{-1} bands in a window around the most flown airspeed, found from the data for each class: 32–40 km h^{-1} for EN B and C, 34–48 km h^{-1} for EN D. The typical 95 % interval at 40 km h^{-1} is $\pm 0.03 \text{ m s}^{-1}$. Above the window the gliders appear to sink less than the curve predicts, most in the first minute after a thermal, although pilots fly faster where the air sinks. The polars use no glide outside the window. In a natural experiment, the sink of the competition glider Enzo 3 drops by 0.03 m s^{-1} at 35 km h^{-1} and 0.07 m s^{-1} at 45 km h^{-1} when a faired race harness comes into use, as a lower drag requires. With the polars removed, 2.3 million glides yield a climatological flow field of the air vertical velocity on a 0.5 km grid. Glides far above the window are rare (6 % of the glide time); without all glides above it, it changes by less than 0.1 m s^{-1} in 95 % of its cells. In years not used to build it, it explains 13 % of the variance of single glides, nearly all that place and hour carry from year to year. Climb rates, in contrast, contain glider and pilot effects of up to 0.35 m s^{-1} . Polars and segmentation code are openly available.

1 Introduction

Thermal convection in the boundary layer is observed by radiosondes, research aircraft, remote sensing and, increasingly, by animals and people who use it. Soaring birds have served as probes of updrafts for decades (Shannon et al., 2002; Treep et al., 2016), and the flight logs of paraglider pilots have been used to compare human and bird soaring strategies (Ákos et al., 2008). Recently, Hernández-Aguayo et al. (2026) assembled 1.47 million climbing segments from 110 730 paraglider flights over France and related three climb observables (thermal ceiling, mean climb rate and its variability) to 120 predictors from the ERA5 reanalysis, aggregated to reanalysis grid cells of 0.25° and one hour. Their flight phases are detected with the method of Vilpellet et al. (2026), which distinguishes search, climb and transition; the analysis of Hernández-Aguayo et al. (2026) uses the climbing phases only.

A variometer does not measure the air. It measures the vertical speed of the glider, which

is the vertical velocity of the air plus the sink of the glider through the air. The second term depends on the glider and its airspeed; the curve that describes it is the sink polar. Hernández-Aguayo et al. (2026) write the climb rate as the thermal updraft minus a glider sink of about $1.0\text{--}1.2\text{ m s}^{-1}$ common to all gliders. This is a fair approximation for a climatology of climbs, but it does not separate glider and air, and it leaves the glides unused: the long straight stretches between thermals, where the air between the updrafts can be observed. To read the air from a glide, the sink of the glider at the airspeed flown has to be known to a few hundredths of a metre per second.

For paragliders such polars are generally not available. Certification according to EN 926-2 classifies gliders by their flight safety characteristics, not by their performance, and manufacturers rarely publish sink polars. Methods to determine polars in flight exist for sailplanes (Marsden, 1985; Wagner, 1994) and have been applied to a paraglider with dedicated instruments (Teskey et al., 2006); wind-tunnel and flight tests address the aerodynamics of paraglider wings (Babinsky, 1999; Kulhánek, 2019) and wind-tunnel tests and simulations that of pilot fairings (Berthus, 2024). To our knowledge, sink polars have not yet been derived from large sets of ordinary flight logs.

This paper does two things. First, it describes a method that measures the sink polar of every glider type with enough recorded flights, from nothing but those flights, and applies it to 101 types (Sections 2–4; all polars are listed and drawn in Appendix B). It rests on one assumption about how pilots use speed. The data show, for each class of gliders, in which window of airspeeds this assumption holds, and the method uses no glide outside it. As a test of its resolution we use a natural experiment, a change of harness among the pilots of one competition glider. Second, it uses the polars to build a flow field of the Valais: the vertical velocity of the air along the glides, resolved in space, height and time of day, and tested on years not used to build it (Section 5). The flow field is a climatological mean; the thermals, which it does not contain, are shown from the climbs at one example place. Section 6 shows what climb rates contain besides the air, in relation to Hernández-Aguayo et al. (2026), and Section 7 discusses the limitations.

2 Data

2.1 Flight logs and segmentation

XContest, an online cross-country contest to which pilots upload their flight logs (XContest, 2026), provided for this study 86 369 IGC flight logs from the Valais, recorded between 2007 and 2026, of which 86 190 could be read. A log stores position and altitude, in most cases once per second, and a free-text glider name in its header. Pilot names were not used.

Each log was processed by the segmenter `flightstates` (version 2.2) and the histogram code `polarmaker` (version 1.4) (Korner, 2026). The log is resampled to 1 s. A second counts as circling if the heading changes by more than 60° within a centred 20 s window; on hand-marked flights this rule agreed with the manual classification for 94.7 % of the seconds. The wind is taken from the drift of each circling phase of at least 1.5 turns and 30 s, and interpolated in time between circling phases. The airspeed of a straight second is the ground velocity minus this wind. Straight flight is cut into segments where the recorded vertical speed or the ground track changes. `polarmaker` stores, for every flight, histograms of the vertical speed of straight seconds in airspeed bands centred on 25, 30, $\dots$, 65 km h^{-1} , after rescaling to a reference altitude (Section 3.2); for the polars the same histograms are computed from the logs in bands 1 km h^{-1} wide, from 25 to 62 km h^{-1} .

2.2 Thermal strength of the day

For every flight with at least 5 min of circling and 5 min of straight flight (77 570 flights) we compute the mean climb rate while circling, as height gained over time spent circling. The thermal strength of a day, W , is the median of these climb rates over all such flights of that day, whatever the glider; it is defined for days with at least five such flights (2 365 days, 66 083 flights). On weak days W is around 0.4 m s^{-1} , on strong days 1.3 m s^{-1} .

2.3 Glider types

Glider names are typed by pilots and come in many spellings. We map them to types with a rule table: manufacturer names, start numbers, colours and sizes are removed, spellings are merged (“alpina4” and “Alpina 4”), and a short list of explicit corrections handles typing errors. Where it is unclear whether two names denote the same wing, they are kept apart: a wrongly separated name costs a few flights, a wrongly merged one biases a polar. The original Zeolite, the Zeolite GT and the Zeolite 2 GT are therefore separate types, as are the Peak 6 and the older Icepeak 6. The Zeolite 2 and the Zeolite 2 GT are the same wing in different cloth and are merged. Table 1 shows how many flights remain at each step. Each type with at least 100 flights on days with known W is a candidate for a polar; there are 106 such types with 26 095 flights, and 101 of them obtain a polar (Section 3.6). The certification class of each type (EN A to D, CCC for competition gliders) was taken from the manufacturers’ documents and test reports; for older wings the LTF class is converted (1-2 as B, 2 as C, 2-3 as D), and for wings certified in different classes by size or weight we use the class of the main sizes at the recommended weight.

Table 1: From flight logs to polars.

Step	Flights
Flight logs provided	86 369
readable	86 190
with at least 5 min of circling and 5 min of straight flight	77 570
on days with a thermal strength W (at least 5 such flights that day; 2 365 days)	66 083
with a glider type from the name table	39 968
in the 106 types with at least 100 flights	26 095
in the 101 types with a polar	25 341
of the 106 types, EN D class (18 types, 1 043 days)	5 096

3 Method: sink polars

3.1 One equation per second

In a second of straight flight the recorded vertical speed $\dot{z}$ is

$$\dot{z} = w + s_g(v), \quad (1)$$

where w is the vertical velocity of the air, v the airspeed and $s_g(v) < 0$ the sink of glider type g in still air, its polar. In any single second w is unknown, so equation (1) cannot be solved for s_g . It can be solved statistically, because $s_g(v)$ is the same in every second flown at the same airspeed, whereas w differs from second to second.

3.2 Density

In thinner air a glider at the same angle of attack flies faster and sinks faster, both by the factor φ^{-1} with

$$\varphi(h) = \sqrt{\rho(h)/\rho(h_0)}, \quad (2)$$

where ρ is the density of the ICAO standard atmosphere (Int, 1993) at the pressure altitude h of the recorder and $h_0 = 2000$ m is the reference altitude. Every second is rescaled, $v_0 = \varphi v$ and $\dot{z}_0 = \varphi \dot{z}$, before it is counted. Between 1000 and 3000 m the factor changes the sink by about 10 %. All polars in this paper refer to 2000 m.

3.3 How pilots use speed

Equation (1) can be solved statistically only if, at a given airspeed, most seconds are flown in air whose vertical velocity does not depend on the choice of that airspeed. Pilots, however, choose their speed according to the air (MacCready, 1958): they slow down in rising air and speed up in sinking air. The method therefore rests on an assumption about this choice: at the airspeeds that the pilots of a class fly most, with little or no speed bar, they glide through ordinary air. Clearly slower and clearly faster flight is kept for special air: pilots slow down in lift, and they fly fast to cross strong sink and while leaving a thermal, where the air still rises. Where the assumption holds, the data follow the physical form of the polar. We therefore let the data decide, for each class, in which window of airspeeds it holds, and determine the polars in this window only (Section 3.6). Of the glide time of all types, 61 % lies in the window of the type, 19 % below and 19 % above it. Fast glides are rare: 6 % of the glide time is flown more than 5 km h^{-1} above the window, 8 % above 47 km h^{-1} . Section 4.2 shows that the special air is visible in the data; the

polars use no glide outside the window, and the flow field hardly changes without the glides above it (Section 5.4).

3.4 The most frequent value

For each glider type and airspeed band k (1 km h^{-1} wide, centred on $v_k = 25, 26, \dots, 62 \text{ km h}^{-1}$) we take the most frequent value of $\dot{z}_0$ over all straight seconds of all flights of that type,

$$m_g(v_k) = \text{mode}\{\dot{z}_{0,i} : |v_{0,i} - v_k| < 0.5 \text{ km h}^{-1}\}, \quad (3)$$

from a histogram with 0.05 m s^{-1} bins, smoothed over five bins and refined by a parabola through the peak. For tables and figures we also use bands 5 km h^{-1} wide, centred on $35, 40, \dots, 60 \text{ km h}^{-1}$.

Figure 1 shows why the mode and not the mean is used. The distribution of the vertical speed at 45 km h^{-1} for the EN D class has a sharp peak and a long tail towards rising air. Mean and median are pulled into the tail, by an amount that differs between airspeeds, because the air a pilot meets depends on how fast he chooses to fly (MacCready, 1958); they describe the pilots' choices as much as the glider. The mode stays where most seconds are. As a control we use the seconds in which the vario was steady for a full minute and no circling occurred within two minutes, selected without looking at the value of the vario. Their mode lies within 0.01 m s^{-1} of the mode of all seconds (Figure 1).

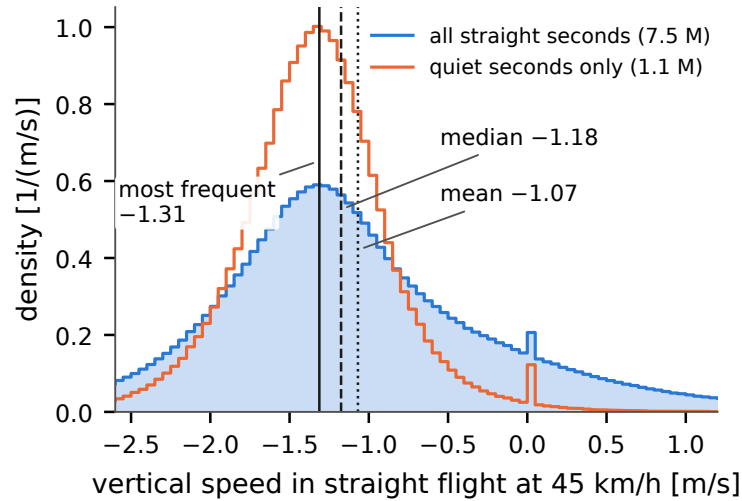


Figure 1: Distribution of the vertical speed in straight flight at $45 \pm 2.5 \text{ km h}^{-1}$ for the EN D class (5096 flights, rescaled to 2000 m, before the subsidence correction). Blue: all 7.5 million straight seconds; orange: the 1.1 million quiet seconds (steady vario for one minute, no circling within two minutes). The most frequent value is -1.31 m s^{-1} for all seconds and -1.32 m s^{-1} for the quiet ones; the median (-1.18) and the mean (-1.07) are pulled towards rising air. The spike at 0 m s^{-1} comes from seconds with unchanged recorded altitude.

3.5 The air between the thermals sinks

The mode in equation (3) is the sink of the glider in the air where most glide seconds are spent, and that air is not still. What rises in the thermals must sink elsewhere. If thermals cover a fraction f of the area and rise at a speed measured by W (Section 3.8 discusses this), mass continuity requires the remaining air to sink on average at

$$\bar{w}_{\text{env}} = -\frac{f}{1-f} W. \quad (4)$$

Pilots, however, do not glide through the sky at random; they choose their lines, and the seconds we count are the seconds they chose. The sink in the air they glide through need not equal $\bar{w}_{\text{env}}$. What equation (4) suggests is the form, a sink proportional to W . We therefore write the mode on a day of thermal strength W as

$$m_g(v_k; W) = s_g(v_k) - c W, \quad (5)$$

with a coefficient c that the flights must determine, and check whether the proportionality holds.

The two unknowns differ in nature. s_g belongs to the glider and is fixed by the position of the peak, which the seconds of one type determine well. c belongs to the air and is seen only across days, because all seconds of a day share one W ; the total shift from the weakest to the strongest days is about 0.15 m s^{-1} , comparable to the uncertainty of a single type on a single day. We therefore determine c once, from the 18 types of the EN D class, a large group of gliders of similar performance, and then the polar of every type with c fixed.

For each EN D type g and each class j of days (W in 0–0.6, 0.6–0.8, 0.8–1.0, 1.0–1.2, 1.2–1.5 m s^{-1}), we compute the deviation $\delta_{g,j}(v_k) = m_{g,j}(v_k) - m_g(v_k)$ of the mode on those days from the mode of the same type on all days. Comparing each type only with itself removes the differences between gliders. The deviations are averaged over the types, weighted by the number of flights in each cell (the variance of the mode of a cell with n flights decreases roughly as $1/n$), and a straight line in W with a common slope and one intercept per group is fitted to three groups of the 1 km h^{-1} bands, 35–38, 39–41 and 42–45 km h^{-1} , weighted by the inverse variance of the class means. Intervals come from resampling the types.

Figure 2 shows the result. The points lie on straight lines: on weak days ($W = 0.35 \text{ m s}^{-1}$) the gliders appear to sink $0.03\text{--}0.05 \text{ m s}^{-1}$ less than on average, on the strongest days ($W = 1.29 \text{ m s}^{-1}$) $0.08\text{--}0.12 \text{ m s}^{-1}$ more. The common slope is $c = 0.118$ (95 %: $0.080\text{--}0.137$). The slopes of the three groups, 0.094 ($0.028\text{--}0.120$), 0.130 ($0.081\text{--}0.155$) and 0.137 ($0.106\text{--}0.169$), agree within their intervals, as they must if the shift is sinking air, which carries a glider down by the same amount at every airspeed. Determined from one type alone, c ranges from -0.03 to 0.24 , which is why it is taken from the class.

Read as a mass balance, with W taken as a measure of the updraft speed, $c = f/(1 - f) = 0.118$ would correspond to thermals covering 11 % of the area, of the size found for thermal plumes in large-eddy simulations and observations of the convective boundary layer, $0.1\text{--}0.2$ (Couvreur et al., 2010). We do not interpret c this way, because it measures the sink in the air the pilots chose, not the mean over the sky. For the polar the distinction does not matter: the polar is counted from the same chosen seconds, and what must be removed from them is the sink in those seconds, which is what c measures.

Every straight second of a flight is therefore raised by $c W$ of its day before it is counted. A flight on a day with $W = 0.7 \text{ m s}^{-1}$ gains 0.08 m s^{-1} , one on a day with $W = 1.3 \text{ m s}^{-1}$ gains 0.15 m s^{-1} . The result is the polar extrapolated to $W = 0$, which we call the polar in still air; Section 3.8 discusses what this limit means.

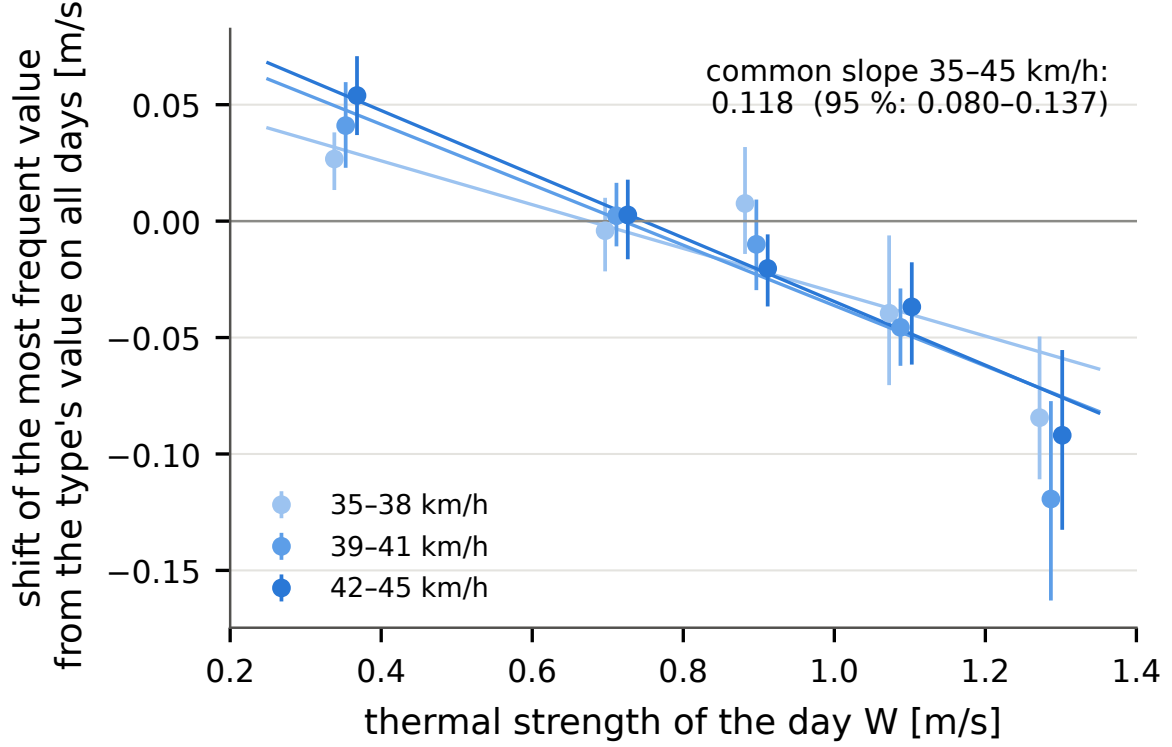


Figure 2: Shift of the most frequent value with the thermal strength of the day, EN D class (18 types), for three groups of airspeed bands. Each point is the flight-weighted mean over the types of the deviation from the type's mode on all days, at the mean W of its class of days; bars are 95 % intervals from resampling types; lines have the common slope $c = 0.118$.

3.6 The physical curve

For a wing at constant loading, a parabolic drag polar $C_D = C_{D0} + KC_L^2$ (Anderson, 2016) gives the sink

$$s(v) = av^3 + \frac{b}{v}, \quad a = -\frac{\rho SC_{D0}}{2mg}, \quad b = -\frac{2Kmg}{\rho S}, \quad (6)$$

with v in m s^{-1} , S the wing area, m the total mass and g the gravitational acceleration. The first term is the drag of glider and pilot, the second the drag due to lift. We fit a and b by weighted least squares to the corrected modes in the 1 km h^{-1} bands of a window of airspeeds. Each band is weighted with the inverse variance of its mode, from resampling flights (Section 3.7) combined with 0.01 m s^{-1} for the resolution of the mode, and is used if it holds at least 3 000 seconds.

The window is found from the data, once for each class. It starts at the most flown airspeed of the class, $\pm 3 \text{ km h}^{-1}$, and grows band by band, downwards and upwards, as long as the new band lies within two standard errors of the curve fitted without it and χ^2 of the whole window stays below its 95 % limit. It starts there because that is where the assumption of Section 3.3 places the ordinary air; the points in special air may also lie on a smooth curve. The windows are $29\text{--}37 \text{ km h}^{-1}$ for EN A (most flown airspeed 34 km h^{-1}), $32\text{--}40 \text{ km h}^{-1}$ for EN B (36) and EN C (37), $34\text{--}48 \text{ km h}^{-1}$ for EN D (39) and $34\text{--}51 \text{ km h}^{-1}$ for the competition class (41). Figure 3 shows them with the points of each class. A type is given the window of its class, shifted by the difference between its own most flown airspeed and that of its class, rounded to 1 km h^{-1} . It obtains a polar if at least eight bands of its window around its most flown airspeed are usable and χ^2 stays below the 95 % limit. This holds for 101 of the 106 types; for four the points do not follow the physical curve within the window, and one has too few seconds (Appendix B). Of the

1 020 band values of the 101 types, 98.8 % lie within their 95 % interval of the fitted curve, and the median of χ^2 per degree of freedom over the types is 0.81. Within the window the curve is a fit to measurements; outside, it is the extension of the physical form. How much of its shape the measurements fix depends on the width of the window. Once the sink s and its slope s' at one airspeed are given, equation (6) fixes the bending, $v^2 s'' = 3s + vs'$. In the narrow windows of the EN A, B and C classes the points determine the level and the slope of the polar around the most flown airspeed; its bending, and with it the airspeed of the best glide ratio and the extension, come from the physical form. In all classes the points lie slightly flatter than the curve: relative to the curve, they are higher at the ends of the window than in its middle, by $0.013\text{--}0.016\text{ m s}^{-1}$ and for EN A by 0.03 m s^{-1} (Figure 3). In the narrow windows this is nearly the whole bending of the curve, and the points of the EN A, B and C classes lie almost on a straight line (Section 7). From equation (6), the best glide ratio is reached at $v^* = (b/a)^{1/4}$ and the minimum sink at $(b/3a)^{1/4}$; both may lie outside the window.

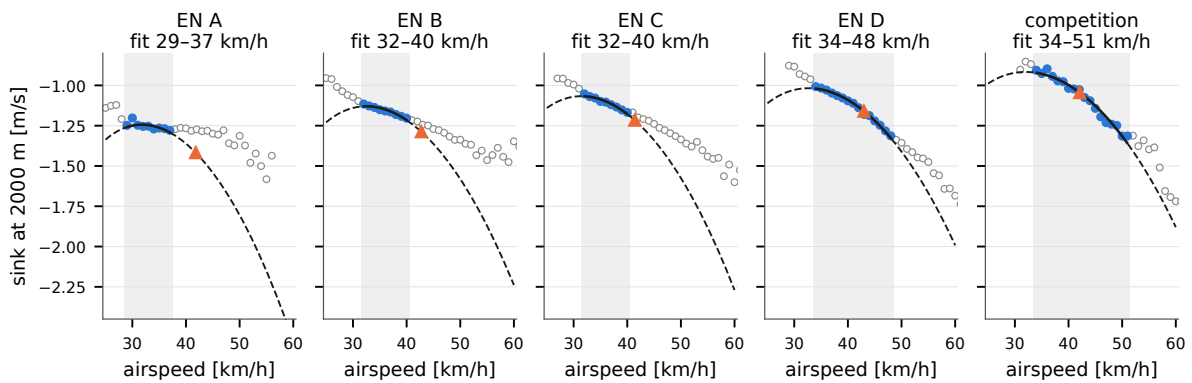


Figure 3: Polars of the five classes, all types of a class together, at 2000 m. Points: most frequent value in 1 km h^{-1} bands after the subsidence correction; filled in the window (shaded), open outside. Line: the physical curve fitted in the window, dashed outside. Triangle: best glide ratio of the curve. Outside the window the points lie above the curve, the gliders appear to sink less (Section 4.2).

3.7 Uncertainty

Seconds within one flight share the air, the trim, the pilot and the wind error and are strongly correlated; resampling seconds would give intervals that are far too narrow. The resampling unit is therefore the flight: flights are drawn with replacement, their histograms summed, the modes and the curve recomputed, 300 times. All 95 % intervals for polars in this paper are percentiles of these repetitions.

3.8 The absolute level

The limit $W = 0$ of equation (5) is not a day we observe. W is the climb rate of circling gliders, that is, the updraft speed minus the sink of a glider in circling flight, about 1 m s^{-1} . If the subsidence is proportional to the updraft speed rather than to the climb rate, it does not vanish at $W = 0$ but amounts to about $c \times 1\text{ m s}^{-1} \approx 0.1\text{ m s}^{-1}$, and all polars are then too pessimistic by up to this amount. Such a constant would shift all polars equally; no comparison between gliders would change. To check the absolute level independently, we use flights without any thermal: at most 60 s of circling, less than 50 m of height gained, 4–40 min long. There are only 63 such flights with a polar (8.5 glide hours) in a cross-country data set. The mean difference between their vario and their polar is $+0.04\text{ m s}^{-1}$ (95 %: -0.06 to $+0.17$; median -0.02). The polars are thus consistent with these flights, and the check is compatible both with no offset

and with an offset of the size just estimated; it cannot exclude a shift of up to 0.17 m s^{-1} . Glides started before 9:00, 19 of the 63 flights, may in addition be affected by cold air sinking along the slopes.

4 Results: sink polars

4.1 The EN D class

Figure 4 shows the polar of the EN D class, all 18 types together, fitted in its window from 34 to 48 km h^{-1} : $a = -3.41 \times 10^{-4} \text{ s}^2 \text{ m}^{-2}$, $b = -6.93 \text{ m}^2 \text{ s}^{-2}$. At 45 km h^{-1} (12.5 m s^{-1}) the drag term contributes -0.665 m s^{-1} and the induced term -0.554 m s^{-1} , together -1.219 m s^{-1} (95 %: -1.226 to -1.212). The 15 bands of the window deviate from the curve by at most 0.02 m s^{-1} ($\chi^2 = 7.3$ for 13 degrees of freedom). The best glide ratio of the curve is 10.3 at 43 km h^{-1} .

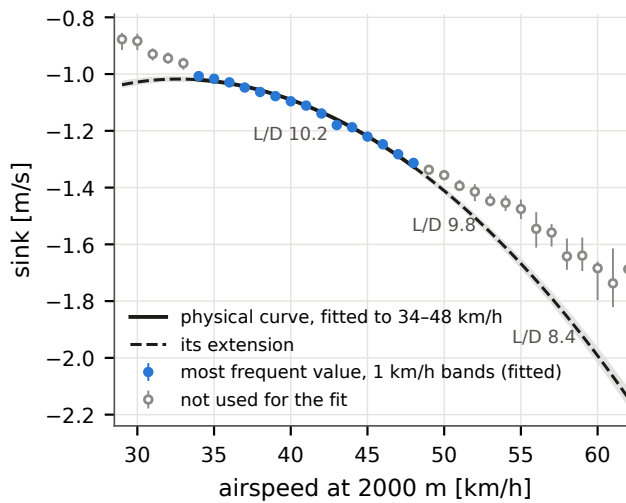


Figure 4: Polar of the EN D class (18 types, 5 096 flights, at 2000 m). Blue: most frequent value in 1 km h^{-1} bands after the subsidence correction, with 95 % intervals from resampling flights. Black: the physical curve fitted to the 15 bands of the window, $34\text{--}48 \text{ km h}^{-1}$, with its 95 % band; dashed: its extension. Open grey circles: bands outside the window; below 34 km h^{-1} and above about 49 km h^{-1} they show less sink than the curve (Section 4.2). L/D: glide ratio of the curve.

4.2 The assumption in the data

Outside the window the gliders appear to sink less than the curve predicts, in all classes and on both sides (Figures 3 and 4). Below the window this is what the assumption expects: pilots fly slowly where the air rises. Above it, the EN D class sinks less than its curve by 0.03 m s^{-1} at 49 km h^{-1} , 0.06 at 50 , 0.19 at 55 and 0.31 m s^{-1} at 60 km h^{-1} . Taken at face value, its measured glide ratio would stay at 10.3 from 45 to 55 km h^{-1} , and that of the EN B class, whose window ends at 40 km h^{-1} , would rise from 9.3 at 40 to 10.7 at 55 km h^{-1} , close to full speed. No paraglider glides better at full speed than near trim: in a flight test of an EN B wing, the Niviuk Hook 3, the best glide ratio, 9.3, was reached at 37.5 km h^{-1} (Heatwole, 2022).

Above the window, the deviation depends on where the fast seconds are flown (Figure 5; for this test each subset of the EN D class has its own curve, fitted to $35\text{--}45 \text{ km h}^{-1}$, within the window). It is largest in the first minute after leaving a thermal, $+0.23 \text{ m s}^{-1}$ at $51\text{--}55 \text{ km h}^{-1}$ (95 %: $0.12\text{--}0.28$), and much smaller more than five minutes from any thermal, $+0.06 \text{ m s}^{-1}$ (-0.01 to 0.16). A wing does not know how long ago its pilot left a thermal; the air does. Pilots

accelerate while the air still rises. Far from thermals a smaller deviation remains above about 54 km h^{-1} .

Pilots also fly faster where the air sinks, as the speed-to-fly rule prescribes (MacCready, 1958). At the places and times where they glide above the window of their type, other pilots, gliding within the window of theirs in the same 1 km cell, 300 m height band and quarter of an hour, find air (computed as in Section 5) that is worse than where pilots glide within the window: by 0.16 m s^{-1} (95 %: $0.14\text{--}0.17$) up to 5 km h^{-1} above the window, 0.22 m s^{-1} at $5\text{--}10 \text{ km h}^{-1}$ above and 0.25 m s^{-1} further above. This acts against the observed deviation: in the most frequent value, the rising air after thermals outweighs it. Below the window the neighbours find better air, by 0.19 m s^{-1} ($0.18\text{--}0.20$) up to 5 km h^{-1} below, as the assumption expects. The choice of speed follows the air gradually, also within the window: where pilots glide in its lower half, the neighbours find air 0.12 m s^{-1} better than where they glide in its upper half (Section 7 discusses what this means for the polars).

Errors of the airspeed, the wind and the vario do not explain the deviation (Appendix A). The seconds outside the window thus lie in special air. None of them enters the polars, and without the glides above the window the flow field hardly changes (Section 5.4).

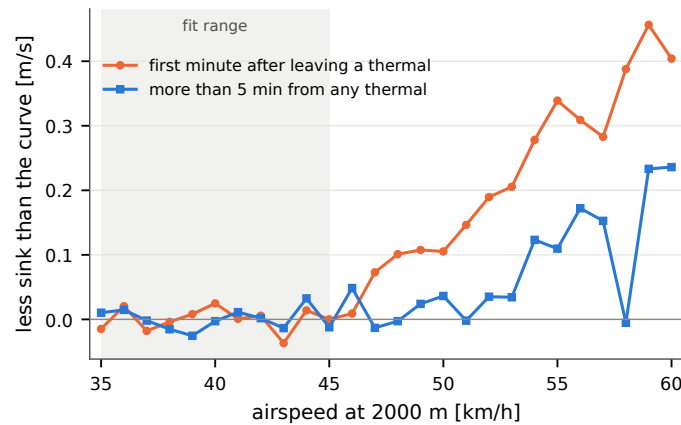


Figure 5: Less sink than the curve fitted to $35\text{--}45 \text{ km h}^{-1}$ (shaded), EN D class, in 1 km h^{-1} bands, for seconds in the first minute after leaving a thermal and for seconds more than five minutes from any thermal; each subset has its own curve.

4.3 All glider types

Table 3 in the appendix lists the polars of all 101 types with their windows and intervals, and Figures 13 and 14 show them with the measured points. The half-width of the 95 % interval at 35 and at 40 km h^{-1} has a median of 0.03 m s^{-1} , from 0.01 for the most-flown types to 0.15 at most, small against the differences between types; polars with wide or very lopsided intervals are marked in the table. The classes separate as expected but overlap: weighted by flights, the sink at 35 km h^{-1} is -1.26 m s^{-1} for the eight EN A types, -1.15 for EN B, -1.09 for EN C, -1.02 for EN D and -0.93 for the competition glider. Several EN C types, among them the Lyght, the Photon and the Artik R, lie close to the EN D class. The curves of the classes (Figure 3) reach their best glide ratio at $41\text{--}43 \text{ km h}^{-1}$: 8.2 for EN A, 9.2 for EN B, 9.5 for EN C, 10.3 for EN D and 11.2 for the competition class. At 2000 m these airspeeds correspond to $37\text{--}39 \text{ km h}^{-1}$ at sea-level density, close to trim speed, as in the flight test of Section 4.2.

4.4 A natural experiment: the Enzo 3 and a faired race harness

A method that measures polars from crowd-sourced flights should be able to detect a known change in a glider–pilot system. The data contain a natural experiment. The Submarine, a race harness with a full fairing around the pilot, was first flown by Ozone team pilots at a Paragliding World Cup Superfinal in 2021 (Ozone Paragliders, 2021; Wentzel, 2024); a review of the harness appeared in January 2024 (Cross Country, 2024). Ozone stated that the shape reduced drag by about 7 %, which corresponds to 0.17 m s^{-1} less sink at full speed than its previous race harness (Wentzel, 2024). Competition pilots flying the Enzo 3, Ozone’s competition glider, are the most likely users. No log records the harness, so the question is whether the polar of the Enzo 3 changes, when, and in what way.

Figure 6a shows the sink of the Enzo 3 at 45 km h^{-1} year by year. 2024 and 2026 lie above all earlier years, 2025 above all except 2020. We therefore compare two periods, to 2023 (435 flights) and from 2024 (155 flights). From 2024 the measured values in bands 5 km h^{-1} wide show less sink by $+0.03 \text{ m s}^{-1}$ at 35 km h^{-1} (95 %: -0.03 to $+0.07$), $+0.05$ at 40 ($+0.01$ to $+0.09$), $+0.09$ at 45 ($+0.03$ to $+0.15$) and $+0.09$ at 50 km h^{-1} ($+0.05$ to $+0.14$). The EN D class changes over the same periods by $+0.04$, $+0.03$, $+0.01$ and -0.03 m s^{-1} , the drift that the air, the pilots and the changing mix of gliders produce in a large group. At 35 and 40 km h^{-1} the change of the Enzo 3 cannot be told from this drift; at 45 and 50 km h^{-1} it is several times larger, and its interval excludes zero.

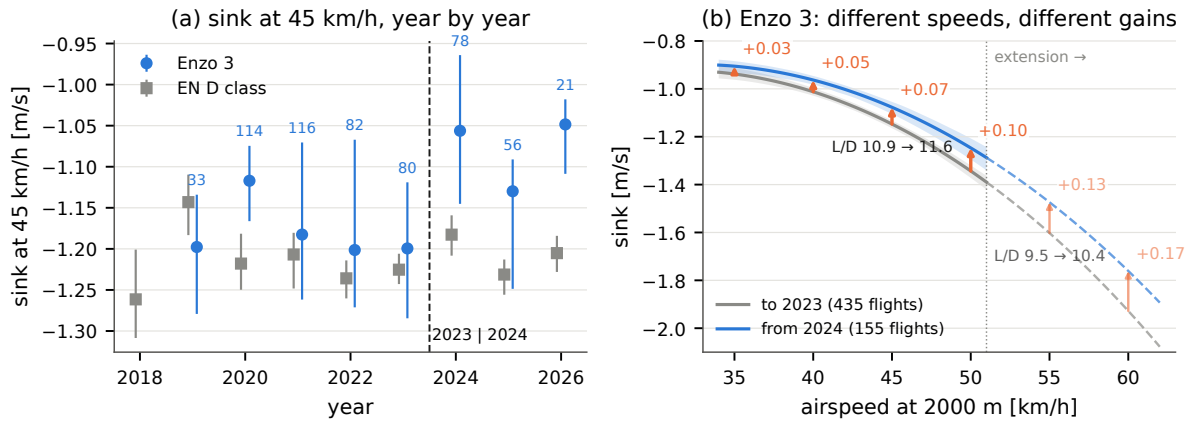


Figure 6: (a) Sink of the Enzo 3 (blue, number of flights above) and of the EN D class (grey) at 45 km h^{-1} per year, with 95 % intervals from resampling flights; the dashed line separates 2023 and 2024. (b) Polars of the Enzo 3 to 2023 (grey, 435 flights) and from 2024 (blue, 155 flights), fitted in the window of the Enzo 3, $34\text{--}51 \text{ km h}^{-1}$, with 95 % bands; dashed above 51 km h^{-1} : extension of the curves. Arrows: gain at each speed, with its value in m s^{-1} ; L/D: glide ratio before and after.

A harness cannot change the wing. It can only change the drag of the pilot, which enters the term av^3 of equation (6); the induced term b/v belongs to the wing. A change of harness must therefore produce a gain that is small at trim speed and grows with speed, and it must lie in a alone. Fitting the two periods in the window of the Enzo 3, $34\text{--}51 \text{ km h}^{-1}$, gives $\Delta a = +3.6 \times 10^{-5} \text{ s}^2 \text{ m}^{-2}$ (95 %: $+0.8$ to $+6.9 \times 10^{-5}$) and $\Delta b = -0.01 \text{ m}^2 \text{ s}^{-2}$ (-0.68 to $+0.45$). The fitted curves give a gain of 0.03 m s^{-1} at 35 km h^{-1} (-0.01 to $+0.06$), 0.07 m s^{-1} at 45 km h^{-1} ($0.04\text{--}0.10$) and 0.10 m s^{-1} at 50 km h^{-1} ($0.05\text{--}0.15$), growing with speed (Figure 6b). At 45 km h^{-1} the drag term accounts for $+0.071 \text{ m s}^{-1}$ of it and the induced term for less than 0.001 m s^{-1} ; this split was not imposed but follows from the bands of each period. The interval of Δa excludes zero; that of Δb corresponds to -0.05 to $+0.04 \text{ m s}^{-1}$ at 45 km h^{-1} . The data thus point to a change of the drag term; a change of the induced term by a few hundredths of a metre per second cannot be excluded. A change in pilot weight would change both terms. A lower drag

of the wing itself, of its lines or its profile, would produce the same signature as a harness and cannot be excluded from the data; the attribution to the harness rests on the timing and on the size of the effect. The extension of the fitted change gives $+0.17 \text{ m s}^{-1}$ at 60 km h^{-1} (95 %: 0.06–0.28), the size of Ozone’s statement for full speed; the comparison is rough, since the reference harnesses differ and the extension is not a measurement. We attribute the change to the harness, with the reservation that this is an attribution, not an observation. Since not all pilots switched at the same time, the measured gain is a lower bound of the gain for a single pilot who switches.

In glide ratio, the Enzo 3 improves from 10.9 to 11.6 at 45 km h^{-1} and from 9.5 to 10.4 at 55 km h^{-1} (the latter from the extension). A single polar for the Enzo 3 would be wrong by about 0.1 m s^{-1} at speed for one of the two periods, so Appendix B gives two.

5 The flow field of the Valais

The flow field built in this section is a climatological mean over two decades of flying days. It describes where and at which hour the air along the glides has tended to rise or sink, not the air of a particular day, and it samples the air along the paths that pilots chose (Section 5.2).

5.1 Construction

With the polars known, every glide segment becomes a measurement of the air. For a straight segment of duration Δt , height change Δh and mean airspeed v at altitude h , flown by a glider of type g ,

$$w = \frac{\Delta h}{\Delta t} - \frac{s_g(\varphi v)}{\varphi}, \quad (7)$$

with $\varphi = \varphi(h)$ from equation (2) and s_g from equation (6); the polar is continued beyond its window, down to 20 and up to 65 km h^{-1} . Glides outside the window rely on this extension, 19 % of the glide time below and 19 % above; if the deviation of Section 4.2 is air, as the special situations suggest, they measure that air. Section 5.4 shows how much the results depend on them. Only flights whose type has a polar are used, and no correction for the day is applied, because the sinking air between the thermals is part of what the flow field describes. We use segments of at least 15 s with φv between 20 and 65 km h^{-1} between 1400 and 4200 m: 2.26 million segments, 42 400 glide hours, 31 952 flights on 2 856 days.

The flow field is the kernel-weighted mean (Nadaraya, 1964)

$$\bar{w}(\mathbf{x}, z, t) = \frac{\sum_s \Delta t_s K_{xy}(\mathbf{x} - \mathbf{x}_s) K_z(z - h_s) K_t(t - t_s) w_s}{\sum_s \Delta t_s K_{xy}(\mathbf{x} - \mathbf{x}_s) K_z(z - h_s) K_t(t - t_s)}, \quad (8)$$

on a grid of 0.5 km cells, with Gaussian kernels of standard deviation 0.75 km in the horizontal, 150 m in height and 45 min in local time, weighted by the duration of each segment. A value is shown only where the denominator, the effective number of glide seconds, is at least 90 s.

5.2 What the flow field shows

Figure 7 shows the flow field at 2500 m at 11:00 and 15:00 local time, for all days. At 11:00 the glides at this height follow the sunlit slopes, and the air along them rises almost everywhere (88 % of 4 502 cells, mean $+0.32 \text{ m s}^{-1}$). At 15:00 the glides cover the whole valley system (11 819 cells). Rising air lies over the slopes and ridges and sinking air over the centre of the Rhône valley and some side valleys (10th percentile -0.13 m s^{-1} , 90th percentile $+0.52 \text{ m s}^{-1}$). This is the pattern of the thermally driven cross-valley circulation, with upslope flow over heated

slopes and compensating subsidence over the valley centre (Zardi and Whiteman, 2013). The enlargement of the central Valais (Figure 8) shows this pattern at the scale of single slopes: closed contours of rising air over the slopes and spurs on both sides of the valley, and the zero line and weakly sinking air near the valley axis.

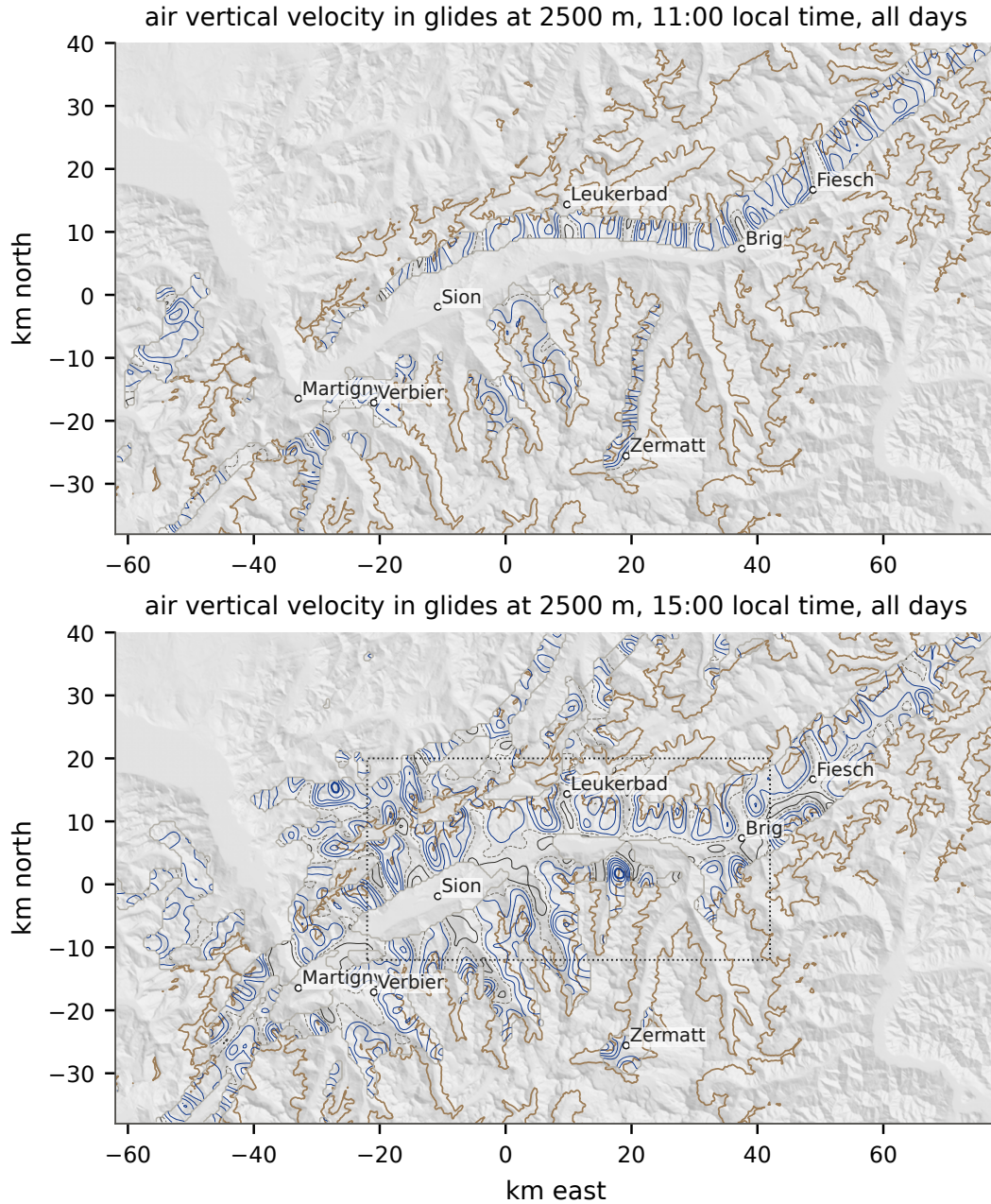


Figure 7: Flow field of the Valais: kernel-weighted mean of the air vertical velocity in glides (equation 8) at 2500 m, all days, at 11:00 (top) and 15:00 (bottom) local time. Contours every 0.2 m s^{-1} : dark blue rising, dark grey sinking, dashed zero. Light grey line: edge of the area with at least 90 effective glide seconds. Brown: the 2500 m terrain contour; shaded relief from the 3'' SRTM elevation model (Farr et al., 2007). Dotted rectangle: area of Figure 8. Coordinates in km from 46.25°N , 7.50°E .

The mean of the flow field is positive, $+0.33 \text{ m s}^{-1}$ over all segments, weighted by duration. This is not a violation of mass continuity, but a property of the sampling. The flow field describes the air along the paths pilots chose, and pilots seek rising air also when they do not circle: along ridges, under cloud streets, in convergence lines. The mode of the air in straight flight is close to

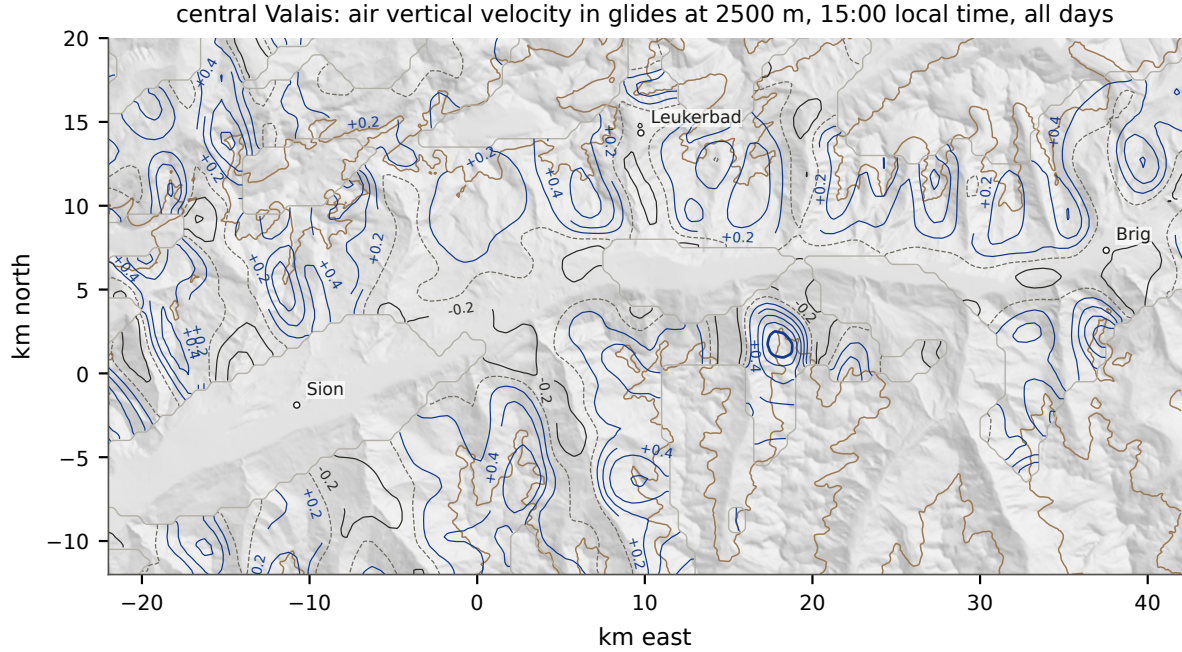


Figure 8: Central Valais, enlarged from Figure 7: air vertical velocity in glides at 2500 m, 15:00 local time, all days. Contours every 0.2 m s^{-1} , labelled in m s^{-1} ; lines and relief as in Figure 7.

zero or slightly negative (Section 3.5); the mean along chosen paths is not. The flow field is thus a map of the air available to a glider that flies like the pilots in the data set, not an areal mean of the vertical velocity. A common offset of all polars (Section 3.8) would shift it by the same amount everywhere.

5.3 The thermals

The flow field contains no circling seconds and therefore no thermal cores. To complete the picture we describe the thermals separately, from the climbs, at one example place. For each climb and 200 m height band we take the time the pilot needs to cross the band while circling, and the positions where he enters and leaves it. On a 1 km grid, per height band and hour ($\pm 1 \text{ h}$), all days together, this gives three quantities. The *frequency* is the fraction of pilots passing through a cell within the band who climb through it there. The *strength* is the median climb rate, corrected with the polars to the sink of the EN D class at 35 km h^{-1} , so that the glider flown does not matter; it remains the climb of an average pilot, $0.15\text{--}0.35 \text{ m s}^{-1}$ below the best quarter (Section 6). The *drift* is the mean horizontal offset between leaving and entering the band, scaled to 1000 m of climb; it shows how the thermals lean.

Figure 9 shows the three quantities with the flow field above Fiesch, one of the most flown places of the Valais. The thermals stand on the sunlit slopes north of the Rhône, between Riederalp and the Eggishorn and above Bellwald. At 11:00 at 2500 m, 41 % of the pilots passing the slope between Fiescheralp and the Eggishorn climb there. The thermals lean up the slope: at 2500 m they drift on average 1.9 km to the north-northwest per 1000 m of climb at 11:00, 1.1 km at 14:00, and 0.5 km to the north-east, up the valley, at 17:00. Their median strength, converted to the EN D class, is 1.8 m s^{-1} at 2500 m and 2.1 m s^{-1} at 3100 m, with little change through the day; what changes is how often they are found. The air in the glides sinks over the mouth of the Fieschertal north of Fiesch, and in the afternoon also over the main valley.

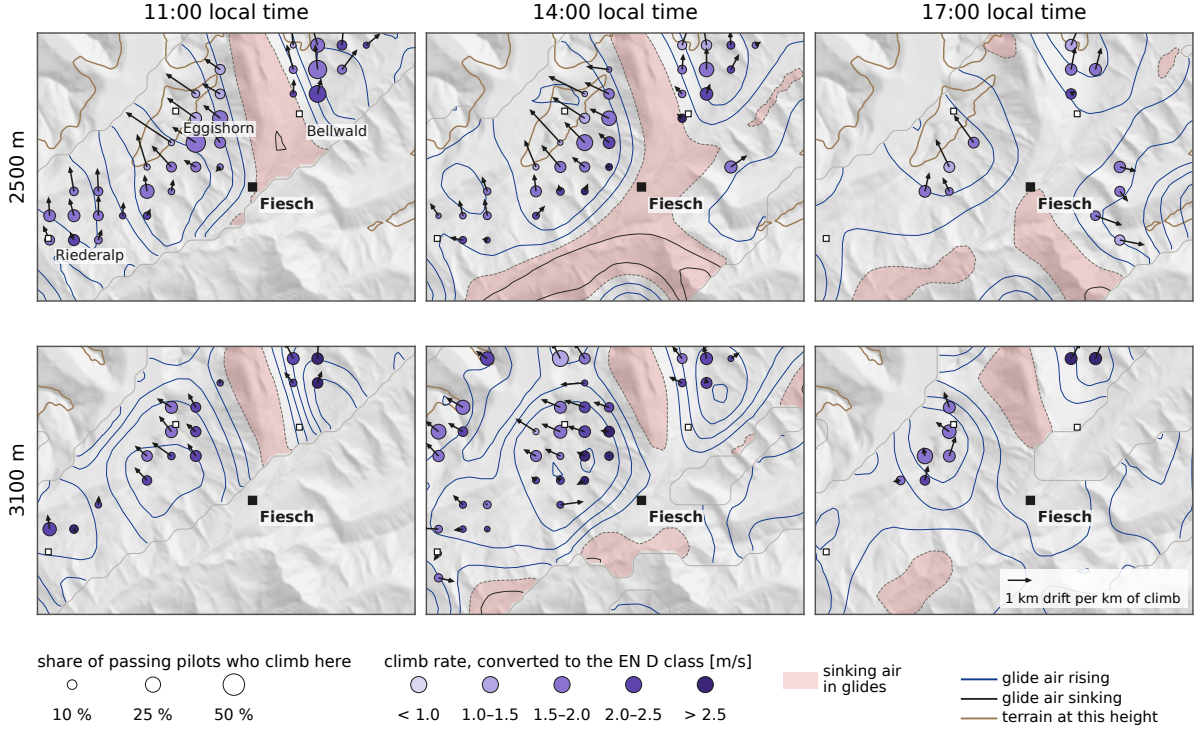


Figure 9: Thermals and flow field above Fiesch, at 2500 m (top) and 3100 m (bottom), at 11:00, 14:00 and 17:00 local time, all days; the area is about 15 by 11 km. Lines: flow field from the glides as in Figure 7, contours every 0.2 m s^{-1} , blue rising, dark grey sinking, shaded where the air sinks. Circles: thermals on a 1 km grid, from the climbs through the 200 m band within $\pm 1 \text{ h}$; area: share of the pilots passing through the cell within the band who climb there; shade: median climb rate, converted to the EN D class with the polars; arrows: mean drift per 1000 m of climb. Only cells with at least 25 passing pilots and five climbs are shown, arrows from ten climbs. Brown: terrain at the height of the map. Squares: Riederalp, Eggishorn, Bellwald and Fiesch.

5.4 A test on years not used

A flow field is useful only if it predicts the air on days that did not contribute to it. We test this by leaving out one year at a time, building the flow field from the remaining years, and predicting the air of every glide segment of the year left out, on days with at least ten flights (1 851 days with usable segments; 2.09 million segments, 39 043 glide hours). For this test the flow field is computed in 0.5 km cells and 400 m height bands. The skill is measured as

$$R^2 = 1 - \frac{\sum_s \Delta t_s (w_s - \hat{w}_s)^2}{\sum_s \Delta t_s (w_s - \bar{w}_{\text{band}})^2}, \quad (9)$$

where $\hat{w}_s$ is the prediction and $\bar{w}_{\text{band}}$ the mean of the training years in the height band of the segment.

A single climatological map explains $R^2 = 0.130$ of the variance of single segments (Figure 10a), 0.11–0.16 below 3000 m and less higher up, where fewer flights contribute. Maps per time of day, rescaled linearly with the thermal strength of the hour (the median climb rate of the circling pilots in that hour of the same day), raise this to 0.136. For means over a day, a 5 km tile and a 400 m band (95 886 groups with at least 300 s of glides), the climatological map explains 0.289 and the time-of-day map 0.308. To judge these numbers we measure how much of the variance is tied to place and time at all. For each segment we draw a partner segment of another flight in the same 1 km cell, 300 m height band and hour, and correlate the deviations of the two from the mean of their height band, weighted by duration. On the same day the two

segments share 34 % of their variance (45 % for segments of at least 60 s); on days of other years they share 14 % (16 %). The climatological map built from other years, with $R^2 = 0.130$, thus captures nearly all that place and time of day carry from year to year; the partner test works on a slightly coarser grid, so the comparison is approximate. A further fifth of the variance is shared by gliders on the same day: the weather of the day, which a map from other years cannot contain. About two thirds are not shared even by two gliders at the same place and hour of the same day: thermals and turbulence on scales below the cell, and measurement noise. The root-mean-square error of a single segment is 0.85 m s^{-1} . The thermal strength of the hour, and in the next version the drift of nearby thermals, are taken from the climbs of the same day: these versions describe the air given what the climbs of that day show, not a forecast.

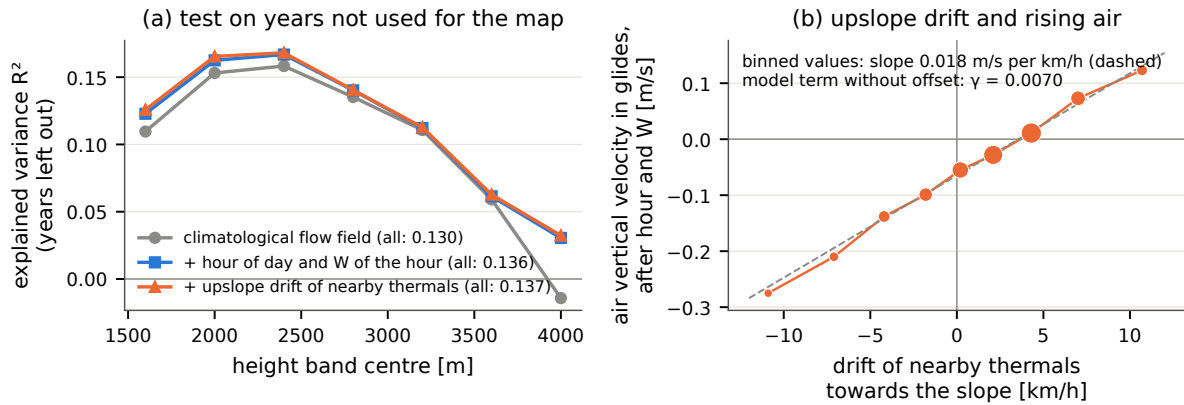


Figure 10: (a) Explained variance of the air vertical velocity of single glide segments in years not used to build the flow field, per height band, for three versions of the flow field: a single climatological map; maps per time of day rescaled with the thermal strength of the hour; and the latter plus a term in the upslope drift of nearby thermals. (b) Air vertical velocity in glides after removing the time-of-day map, against the component of the drift of thermals within about 3 km, 300 m and 1 h on the same day directed up the local slope; marker area grows with glide hours; dashed: weighted linear fit to the binned values.

The drift of thermals near a glide, their horizontal speed from the wind estimate of the circling phases (Section 2), carries information about the local circulation. Figure 10b shows the air vertical velocity in glides, after removing the time-of-day map, against the component of this drift towards the local slope. The relation is close to linear, from -0.28 m s^{-1} at -11 km h^{-1} (drift away from the slope) to $+0.12 \text{ m s}^{-1}$ at $+11 \text{ km h}^{-1}$ (towards it), with a slope of 0.018 m s^{-1} per 1 km h^{-1} (weighted fit to the binned values; offset -0.065 m s^{-1} at zero drift). In the predictive version of the flow field this term was fitted without an offset, $\gamma = 0.0070 \text{ m s}^{-1}$ per 1 km h^{-1} (range 0.0064–0.0079 over the years left out). It raises R^2 for single segments from 0.136 to 0.137 and that of day-and-tile means from 0.308 to 0.311: large drifts are rare, and part of the local circulation is already contained in the map.

The flow field hardly depends on the glides above the window. Without them, 19 % of the glide time, the maps of Figure 7 change by less than 0.10 m s^{-1} in 95 % of the cells (correlation 0.99), most of it a common shift of $+0.04 \text{ m s}^{-1}$, since fast glides lie in sinking air; R^2 changes from 0.130–0.137 to 0.127–0.133. The glides below the window lie in rising air: without them as well, the maps keep their pattern (correlation 0.98) but shift by -0.06 m s^{-1} .

6 What climbs measure

Hernández-Aguayo et al. (2026) and this study use the same kind of data for different questions. Theirs covers a whole country at the scale of a reanalysis cell and one hour and asks which meteorological variables control the thermals; boundary-layer height emerges as the leading

predictor of ceiling and climb rate. Ours covers one valley system on a 0.5 km grid and asks how the air moves along the glides. Both read the air from the motion of gliders, and both need to know what else that motion contains.

A climb rate is the vertical velocity of the air on the pilot's circle minus the sink of the glider in the turn. Hernández-Aguayo et al. (2026) take this sink as $1.0\text{--}1.2\text{ m s}^{-1}$ for all gliders and the mean climb rate of all pilots as the thermal strength. On average the sink is about right: from the polars and the turn times below, the median sink in the turn is 1.16 m s^{-1} . But a climb rate contains four quantities besides the air, and they differ between gliders, heights and pilots (Figure 11a).

1. *The glider.* At 35 km h^{-1} , the closest of our bands to circling speed, the flight-weighted class means differ by 0.13 m s^{-1} between EN B and EN D and by 0.33 m s^{-1} between EN A and the competition glider (Section 4.3).
2. *The height.* In thinner air a glider sinks faster (Section 3.2): the EN D class by 0.05 m s^{-1} more at 2000 m than at 1000 m, and by 0.10 m s^{-1} more at 3000 m.
3. *The bank angle.* The segmented logs give the duration and the number of turns of each circling piece, hence the turn time T . With the airspeed v of minimum sink from the polar, $2\pi/T = g \tan \beta / v$ gives the bank angle β ; at constant angle of attack the sink grows by $1/\cos^{3/2} \beta$. The median turn takes 27 s, with a radius of 40 m and a bank of 13° ; the turn adds $0.03\text{--}0.06\text{ m s}^{-1}$ (quartiles) to the sink. Irregular circles, while a pilot centres, lengthen the turn time, so this is a lower bound.
4. *The pilot.* Pilots centre a thermal with different skill. In climbs shared by at least three pilots we compare, for each pilot and 200 m height band, the time needed to cross the band while circling with that of the other pilots crossing the same band within 300 m and 5 min (108 600 crossings on 966 days). To separate skill from chance, pilots are selected in one band and measured in the next band of the same climb. The upper quarter then climbs 0.35 m s^{-1} (95 %: $0.33\text{--}0.36$) faster than the others, the best tenth 0.53 m s^{-1} ($0.50\text{--}0.55$). Within 150 m and 1 min, where pilots share the same part of the thermal at the same moment and not different pulses of it, the advantage is 0.15 m s^{-1} ($0.12\text{--}0.18$) and 0.24 m s^{-1} ($0.20\text{--}0.30$). Removing the sink of the gliders with the polars changes none of these values by more than 0.01 m s^{-1} . The advantage is smaller in gaggles of more than ten pilots, presumably because the others mark the core.

Glider, height and bank angle can be removed with the polars, the density and the turn time; a climb then becomes a measurement of the air on the pilot's circle. The pilot can only be bounded. A fifth property cannot be removed: pilots circle only where the air rises faster than their glider sinks in the turn, so the climbs sample the upper tail of the air. Along the glides the air rises faster than 1.16 m s^{-1} during 16 % of the time; in the climbs it rises at 2.2 m s^{-1} on average (Figure 11b).

A constant offset would not change the correlations of Hernández-Aguayo et al. (2026). But gliders, pilots, heights and the selection change with terrain, season and the strength of the day, and the glider and pilot effects are each of the size of the difference they find between plains and high mountains, 1.01 against 1.35 m s^{-1} in the warm season. Part of that difference may belong to gliders and pilots rather than to the air; the height acts the other way, for climbs over high mountains lie higher, where gliders sink more. Our own thermal strength W (Section 2.2) is a climb rate too; it serves only as an index of the day, and only the product cW enters the polars.

The glides add information that the climbs do not contain. They sample the air between the thermals, they show where it rises without circling and where it sinks, and they do so at the resolution of the terrain, with only the polar to remove. Hernández-Aguayo et al. (2026) name

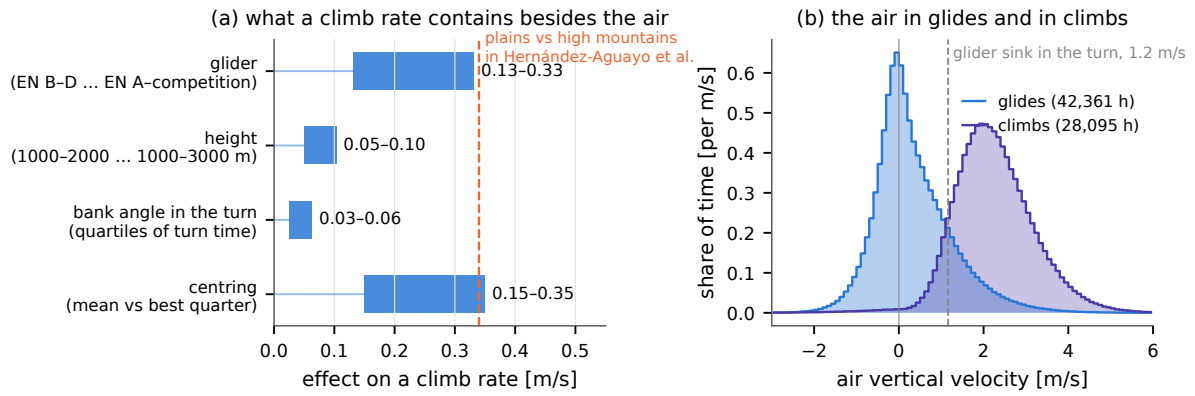


Figure 11: What a climb rate contains besides the air. (a) Size of four effects, each bar from a smaller to a larger comparison: glider (sink at 35 km h^{-1} , flight-weighted class means, EN B against EN D to EN A against the competition glider), height (sink of the EN D class at 1000 m against 2000 to 3000 m), bank angle (additional sink in the turn, quartiles of the turn time), pilot (all others against the best quarter in the same thermal, narrow to wide window). Dashed: difference of the mean climb rate between plains and high mountains in Hernández-Aguayo et al. (2026). (b) Air vertical velocity along the glides and in the climbs (climb rate plus the sink of the glider in the turn, from its polar, the density and the turn time), weighted by time, 1400–4200 m, flights with a type polar. Dashed: median sink of the glider in the turn.

the transition phases as a natural extension of their pipeline, for the spacing of thermals; to read the vertical motion of the air along the glides, the polars are needed.

7 Discussion

The assumption about speed. All quantities in this paper are conditional on where and when pilots fly: on flyable days, along chosen lines, mostly in spring and summer, and at speeds chosen according to the air. The polars rest on the assumption that within the window of its class pilots glide through ordinary air. Within the window it is supported by the bands lying on the two-parameter physical curve, by the quiet seconds giving the same mode as all seconds, and by a subsidence slope that is the same at all airspeeds. It holds only approximately: the choice of speed follows the air gradually, also within the window, where pilots gliding in its lower half find air 0.12 m s^{-1} better than those in its upper half (Section 4.2). As far as this difference reaches the most frequent value, it makes the polars steeper than the gliders are. It cannot reach it in full: the polars would then be nearly flat across the window, and the best glide ratio would lie near full speed instead of near trim (Section 4.3). The data do not tell how much of it reaches the mode; a part of a few hundredths of a metre per second cannot be excluded. Within the window the points also lie slightly flatter than the physical curve, higher at its ends than in its middle (Section 3.6). An error of the airspeed of the single second does not explain this, although such an error moves seconds towards the most flown airspeed: placing every band at the mean airspeed actually flown by its seconds, recovered from the measured distribution of airspeeds and the measured scatter of the single second, 5.1 km h^{-1} , makes the points steeper but leaves them as straight as before. More likely, the special air does not begin abruptly at the edge of the window but grows gradually towards both ends; in addition, below trim speed pilots brake, which changes the shape of the wing. The logs do not separate the two. The effect is about half the 95 % interval of a single type, but for the EN A, B and C classes it means that the shape of the polar, within the window and beyond, rests on the physical form and not on the points. Outside the window the results do not depend on the assumption: the polars use no glide there, and the flow field hardly changes without the glides above it.

What a polar of a type means. A type polar averages over sizes, wing loadings, trims and harnesses as flown in the Valais. Wing loading moves a polar along a line through the origin by the square root of the load factor; we do not know the loads. The harness, as the Enzo 3 shows, changes the drag term measurably. A polar in this paper is therefore the polar of a type as it is flown, not of a type as certified. With a few hundred flights, a type obtains a polar with an interval of a few hundredths of a metre per second within its window. This is enough to follow a model line through its generations (Appendix B lists the types of each model line by year of release): the sink of the Klimber at 35 km h^{-1} decreases from 1.10 m s^{-1} for the Klimber P (2016) to 1.05 m s^{-1} for the Klimber 2 P (2021) and 0.97 m s^{-1} for the Klimber 3 P (2023); part of such a difference may reflect the age of the wings and the harnesses flown with them. Polars of this kind complement test flights and wind-tunnel measurements.

Which flights. Only flights with a recognisable glider name contribute to the polars, and only flights of types with a polar contribute to the flow field: 33 216 of 86 190 flights, 31 952 of them with usable glides, biased towards recent gliders. Outside its window a polar is an extension of the physical form, not a measurement. For the EN A, B and C types the window ends between 35 and 43 km h^{-1} , so their polars with the speed bar pushed need an independent measurement, such as flights with a calibrated airspeed sensor in still air. The absolute level rests on the still-air limit and on 63 flights without thermals.

Beyond one valley. The method needs nothing but enough logs per glider type. In other regions some assumptions will be tested harder than in the Valais. Where glides go mostly in one direction, a wind error no longer cancels between headwind and tailwind, and the polar

should be computed separately for both, which must agree if the wind is right. Where thermals are much stronger, the proportionality of equation (5) must be checked before it is extrapolated, and the thermal strength of a day has to be defined for a region rather than a valley. Glider names in other languages will need more care.

8 Conclusions

Sink polars of paragliders can be measured from ordinary flight logs. At fixed airspeed the glider is a constant and the air is not, so the most frequent vertical speed in straight flight estimates the sink; corrected for air density and for the subsidence of the glide air, and fitted with the physical form of the polar in 1 km h^{-1} bands, it gives polars with typical 95 % intervals of $\pm 0.03 \text{ m s}^{-1}$ at 40 km h^{-1} for 101 types; a common offset of all polars of up to about 0.1 m s^{-1} cannot be excluded. The method rests on an assumption about how pilots use speed: at the airspeeds they fly most, they glide through ordinary air, and they keep other speeds for special air, slow flight for lift, fast flight for strong sink and for the rising air in which they leave a thermal. The data show where the assumption holds: in a window around the most flown airspeed, found for each class, in which the points follow the physical curve, $32\text{--}40 \text{ km h}^{-1}$ for EN B and C and $34\text{--}48 \text{ km h}^{-1}$ for EN D. Outside the window they show the special air. The polars use only the window. Fast glides are rare, and the flow field does not need them: without the glides above the window it changes by less than 0.1 m s^{-1} in 95 % of its cells. The subsidence correction is measured from the flights themselves; it grows linearly with the thermal strength of the day, by 0.12 m s^{-1} per 1 m s^{-1} , and is the same at all airspeeds within its uncertainty. In a natural experiment, the method detects a change in the polar of one glider that coincides with the arrival of a faired race harness and grows with speed, as a lower drag requires.

With the polars removed, glide segments become measurements of the air. Over the Valais they show rising air over sunlit slopes and ridges, compensating subsidence over the valley centre in the afternoon, and a dependence on the local drift of thermals. A flow field built from other years explains a modest but robust part of the variance of single glides and about a third of the variance of daily means over 5 km tiles; the rest is the weather of the day and variability below the scale of the maps. Above Fiesch, the climbs add the thermals: where they stand, how often they are found, how strong they are and how they lean. Glider polars are thus the step that turns the glides in flight logs into observations of the air. Climbs, too, become measurements of the air once glider, height and bank angle are removed with the polars, the density and the turn time; the centring of the pilot and the choice of where to circle remain, and the pilot effect alone is of the size of the differences that climb-based studies compare.

The flow field is a climatological mean for one valley system. It can serve as a coarse guide for pilots and as the environment for simulations of cross-country flight, in which synthetic pilots fly a measured polar through the measured air. Extended to the Alpine arc, with the gaps between flown paths filled, such simulations may identify routes that have not yet been flown and quantify what a difference between two polars is worth on a given route. The flow field may also serve as an observational reference for models of slope and valley circulations.

Data and code availability

The flight logs were provided by XContest for this study; they are not public, and pilot identities were not used. The segmentation and histogram code (`flightstates 2.2`, `polarmaker 1.4`) is openly available (Korner, 2026). The polars of Appendix B are provided in machine-readable form as an ancillary file to this article. The scripts that compute the polars, the flow field and the figures of this paper are available from the author.

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A Tests of the method

Above the window the gliders appear to sink less than the curve predicts (Section 4.2). We tested the obvious causes of this deviation on the 5 309 flights of the EN D types on days with known W , read directly from the logs with the glider name of the log header (a selection slightly different from Table 1), with the airspeed in 1 km h^{-1} bands from 35 to 60 km h^{-1} and a curve fitted to $35\text{--}45 \text{ km h}^{-1}$ for every variant, a common range inside the window of the class, $34\text{--}48 \text{ km h}^{-1}$, so that the variants can be compared (Figure 12). (i) Errors of the airspeed move seconds from the populated slower bands into the sparse fast ones. The vario is averaged over about 30 s, whereas the airspeed is that of the single second, which scatters by 5.1 km h^{-1} around its 30-s average. Removing this error, by binning the seconds by the airspeed averaged like the vario, or by adding further error of 2, 4 and 6 km h^{-1} and extrapolating back to none (Cook and Stefanski, 1994), shifts all bands but leaves the deviation above 47 km h^{-1} as it is. (ii) The exchange of speed and height when the pilot accelerates or slows down: a total-energy vario, which adds $v \dot{v}/g$, shows the same deviation. (iii) Errors of the wind estimate: the deviation grows with the wind, but it is present when the wind is below 7 km h^{-1} . (iv) Selection of flights: it is present in the quarter of flights with the most seconds above 50 km h^{-1} as in the other flights, and on weak as on strong days. Three further restrictions leave the deviation unchanged within its uncertainty: to seconds with a steady airspeed over the averaging window of the vario and no circling within 30 s, to seconds within two minutes of a thermal, where the wind is measured, and to a vario computed from the GNSS altitude instead of the pressure altitude.

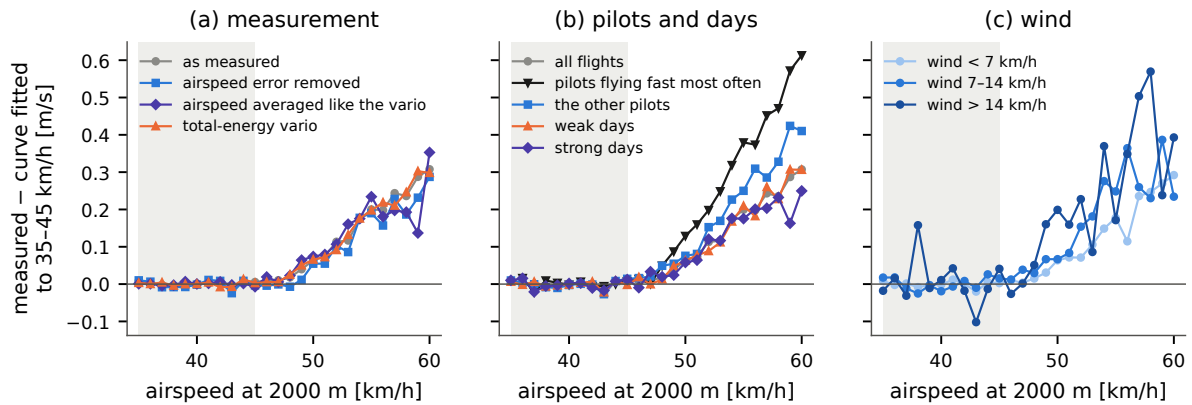


Figure 12: Deviation of the most frequent value from the curve fitted to $35\text{--}45 \text{ km h}^{-1}$ (shaded), EN D class (5 309 flights read from the logs), 1 km h^{-1} bands at 2000 m, after the subsidence correction; positive for less sink. Each variant has its own curve. (a) As measured; with the error of the airspeed removed by extrapolation from added errors of 2, 4 and 6 km h^{-1} ; with the seconds binned by the airspeed averaged like the vario; with a total-energy vario. (b) The quarter of flights with the largest share of seconds above 50 km h^{-1} and the other flights; days with W below and above 0.9 m s^{-1} . (c) Seconds with an estimated wind below 7, between 7 and 14, and above 14 km h^{-1} .

Two choices of the method were tested on all 6 098 flights of the EN D types read directly from the logs, on all days, since no subsidence correction is needed for this test (Table 2). Narrowing the airspeed band from 5 to 1 km h^{-1} changes the mode by at most 0.01 m s^{-1} ; the interval of a single band widens because seconds are lost, but many narrow bands constrain the curve better than a few wide ones. Recomputing the airspeed with half and with one and a half times the measured wind changes the mode by at most 0.02 m s^{-1} between 35 and 45 km h^{-1} and by 0.03 m s^{-1} at 50 km h^{-1} . Glides in the Valais are flown in all directions, so an error of the wind acts in opposite directions on headwind and tailwind seconds and largely cancels.

Table 2: Sensitivity of the mode (m/s, EN D class, 6 098 flights, at 2000 m, before the subsidence correction) to the width of the airspeed band and to a scaling of the wind used to compute the airspeed. In brackets: 95 % interval from resampling flights.

	35 km h ⁻¹	40 km h ⁻¹	45 km h ⁻¹	50 km h ⁻¹
band 1 km h ⁻¹ (fit)	-1.096	-1.173	-1.308	-1.438
band 2 km h ⁻¹	-1.095	-1.174	-1.304	-1.445
band 3 km h ⁻¹	-1.098	-1.173	-1.301	-1.452
band 5 km h ⁻¹ (tables)	-1.097	-1.174	-1.304	-1.447
	(-1.105, -1.089)	(-1.182, -1.167)	(-1.313, -1.294)	(-1.459, -1.431)
band 7 km h ⁻¹	-1.101	-1.176	-1.299	-1.439
wind × 0.5	-1.080	-1.163	-1.296	-1.448
wind × 1.5	-1.115	-1.186	-1.298	-1.417

B Polars of all glider types

Table 3: Sink polars of the 101 glider types with at least 100 flights on days with a known thermal strength whose points follow the physical curve within their window, in still air at 2000 m (ISA density), with the subsidence correction $c = 0.118$ (Section 3.5). Window: airspeed range of the fit (Section 3.6), the window of the class shifted by the difference between the most flown airspeed of the type and that of its class. $s(v)$ is read from the curve $s = av^3 + b/v$; values in brackets lie outside the window and are extensions of the physical form, not measurements. Super- and subscripts give the 95 % interval from resampling flights. [†]: uncertain polar (within the window, an interval wider than 0.2 m s^{-1} , or one side of an interval more than four times the other). $(L/D)_{\max}$ and the minimum sink are properties of the fitted curve, with the airspeed in km/h; speeds marked * lie outside the window and are extrapolated. MS: 99th percentile of the measured airspeed in glides (km/h, at 2000 m). Year: first certification test or, where none was found, the manufacturer's launch; types are sorted by class, within a class by model line, and within a model line by year, so that each line shows its development steps. Class: EN certification class according to the manufacturers and test reports, for older wings the LTF class (1-2 as B, 2 as C, 2-3 as D); for wings certified in different classes by size or weight, the class of the main sizes at the recommended weight; CCC: competition class. Names are the type names as written by pilots, merged by the rules of Section 2.3. Figures 13 and 14 show the polars with the measured points.

Type	Year	Class	Flights	Days	Window km/h	a $10^{-4} \text{ s}^2 \text{ m}^{-2}$	b $\text{m}^2 \text{ s}^{-2}$	$s(35)$ m/s	$s(40)$ m/s	$s(45)$ m/s	$(L/D)_{\text{max}}$ @ km/h	s_{min} m/s @ km/h	MS km/h
Competition class (1 type)													
Enzo 3	2017	CCC	590	263	34–51	−3.272	−6.10	$-0.93^{+0.02}_{-0.02}$	$-1.00^{+0.01}_{-0.01}$	$-1.13^{+0.02}_{-0.01}$	11.2 @ 42	−0.92 @ 32*	61
EN D (17 types)													
Klimber P	2016	D	273	195	33–47	−2.897	−8.05	$-1.09^{+0.04}_{-0.03}$	$-1.12^{+0.03}_{-0.02}$	$-1.21^{+0.04}_{-0.04}$	10.3 @ 46	−1.09 @ 35	54
Klimber 2 P	2021	D	284	164	33–47	−3.158	−7.38	$-1.05^{+0.03}_{-0.03}$	$-1.10^{+0.02}_{-0.03}$	$-1.21^{+0.02}_{-0.03}$	10.4 @ 44	−1.05 @ 34	55
Klimber 3 P	2023	D	186	119	33–47	−3.350	−6.45	$-0.97^{+0.03}_{-0.02}$	$-1.04^{+0.02}_{-0.02}$	$-1.17^{+0.03}_{-0.05}$	10.8 @ 42	−0.96 @ 32*	56
LM 7	2019	D	212	150	32–46	−4.151	−5.86	$-0.98^{+0.02}_{-0.02}$	$-1.10^{+0.02}_{-0.02}$	$-1.28^{+0.03}_{-0.03}$	10.1 @ 39	−0.94 @ 30*	52
Mantra 7	2019	D	255	164	34–48	−2.904	−7.95	$-1.08^{+0.02}_{-0.02}$	$-1.11^{+0.02}_{-0.01}$	$-1.20^{+0.02}_{-0.02}$	10.4 @ 46	−1.08 @ 35	53
Omega 7	2006	D	130	89	32–46	−3.772	−8.02	$-1.17^{+0.03}_{-0.03}$	$-1.24^{+0.03}_{-0.02}$	$-1.38^{+0.06}_{-0.03}$	9.1 @ 44	−1.17 @ 33	52
Omega X-Alps	2015	D	127	76	33–47	−3.566	−7.01	$-1.05^{+0.04}_{-0.03}$	$-1.12^{+0.03}_{-0.02}$	$-1.26^{+0.03}_{-0.03}$	10.0 @ 43	−1.04 @ 32*	56
Omega X-Alps 3	2019	D	250	165	33–47	−3.884	−6.69	$-1.04^{+0.02}_{-0.02}$	$-1.14^{+0.02}_{-0.02}$	$-1.29^{+0.04}_{-0.04}$	9.8 @ 41	−1.02 @ 31*	56
Omega ULS	2023	D	258	151	34–48	−3.403	−7.32	$-1.07^{+0.03}_{-0.03}$	$-1.13^{+0.02}_{-0.02}$	$-1.25^{+0.02}_{-0.02}$	10.0 @ 44	−1.06 @ 33*	54
Peak 3	2013	D	157	111	32–46	−3.488	−7.33	$-1.07^{+0.04}_{-0.03}$	$-1.14^{+0.02}_{-0.02}$	$-1.27^{+0.04}_{-0.04}$	9.9 @ 43	−1.07 @ 33	52
Peak 6	2022	D	180	133	34–48	−3.623	−6.14	$-0.96^{+0.03}_{-0.05}$	$-1.05^{+0.02}_{-0.03}$	$-1.20^{+0.04}_{-0.02}$	10.6 @ 41	−0.94 @ 31*	58
XC Racer	2018	D	112	78	36–50	−3.390	−6.92	$(-1.02^{+0.07}_{-0.06})$	$-1.09^{+0.04}_{-0.05}$	$-1.22^{+0.03}_{-0.03}$	10.3 @ 43	−1.02 @ 33*	59
Zeno	2016	D	453	208	34–48	−3.199	−6.72	$-0.98^{+0.02}_{-0.02}$	$-1.04^{+0.01}_{-0.02}$	$-1.16^{+0.02}_{-0.02}$	10.8 @ 43	−0.98 @ 33*	57
Zeno 2	2022	D	696	225	37–51	−3.417	−6.40	$(-0.97^{+0.03}_{-0.02})$	$-1.04^{+0.02}_{-0.01}$	$-1.18^{+0.02}_{-0.01}$	10.7 @ 42	−0.96 @ 32*	60
Zeolite	2019	D	234	169	34–48	−3.554	−6.97	$-1.04^{+0.04}_{-0.05}$	$-1.11^{+0.03}_{-0.03}$	$-1.25^{+0.03}_{-0.02}$	10.1 @ 43	−1.03 @ 32*	57
Zeolite GT	2019	D	508	249	34–48	−3.593	−6.90	$-1.04^{+0.03}_{-0.02}$	$-1.11^{+0.01}_{-0.01}$	$-1.25^{+0.02}_{-0.02}$	10.0 @ 42	−1.03 @ 32*	57
Zeolite 2 GT	2023	D	637	210	35–49	−3.888	−6.02	$-0.98^{+0.02}_{-0.02}$	$-1.07^{+0.01}_{-0.01}$	$-1.24^{+0.02}_{-0.02}$	10.3 @ 40	−0.95 @ 30*	58
EN C (27 types)													

Table 3 continued

Type	Year	Class	Flights	Days	Window	a	b	$s(35)$	$s(40)$	$s(45)$	$(L/D)_{\max}$	$s_{\min}$	MS
Alpina 2	2013	C	127	86	31–39	-4.581	-6.41	$-1.08^{+0.04}_{-0.03}$	$(-1.21^{+0.05}_{-0.05})$	$(-1.41^{+0.11}_{-0.11})$	9.2 @ 39*	-1.03 @ 30*	52
Alpina 3	2017	C	303	173	31–39	-4.404	-6.37	$-1.06^{+0.02}_{-0.01}$	$(-1.18^{+0.03}_{-0.02})$	$(-1.37^{+0.06}_{-0.05})$	9.4 @ 40*	-1.02 @ 30*	52
Alpina 4	2020	C	871	360	31–39	-3.838	-7.07	$-1.08^{+0.01}_{-0.01}$	$(-1.16^{+0.02}_{-0.02})$	$(-1.31^{+0.03}_{-0.03})$	9.6 @ 42*	-1.06 @ 32	52
Artik 6	2020	C	334	210	32–40	-4.415	-6.30	$-1.05^{+0.02}_{-0.03}$	$-1.17^{+0.05}_{-0.03}$	$(-1.37^{+0.10}_{-0.07})$	9.5 @ 39	-1.01 @ 30*	52
Artik R	2022	C	279	162	33–41	-3.927	-6.66	$-1.05^{+0.03}_{-0.02}$	$-1.14^{+0.03}_{-0.03}$	$(-1.30^{+0.07}_{-0.08})$	9.8 @ 41*	-1.02 @ 31*	54
Artik 7 P	2024	C	299	127	33–41	-3.722	-6.88	$-1.05^{+0.02}_{-0.02}$	$-1.13^{+0.03}_{-0.02}$	$(-1.28^{+0.05}_{-0.05})$	9.9 @ 42*	-1.03 @ 32*	53
Bonanza 2	2018	C	106	93	32–40	-3.345	-7.77	$-1.11^{+0.04}_{-0.02}$	$-1.16^{+0.04}_{-0.05}$	$(-1.27^{+0.06}_{-0.10})$	9.8 @ 44*	-1.10 @ 34	55
Bonanza 3	2023	C	109	81	33–41	-3.125	-7.54	$-1.06^{+0.05}_{-0.04}$	$-1.11^{+0.03}_{-0.05}$	$(-1.21^{+0.07}_{-0.13})$	10.3 @ 45*	-1.06 @ 34	54
Camino	2019	C	129	99	34–42	-3.721	-7.69	$-1.13^{+0.04}_{-0.04}$	$-1.20^{+0.03}_{-0.05}$	$(-1.34^{+0.08}_{-0.12})$	9.3 @ 43*	-1.12 @ 33*	53
Cure	2015	C	106	93	32–40	-4.009	-7.38	$-1.13^{+0.05}_{-0.05}$	$-1.21^{+0.04}_{-0.03}$	$(-1.37^{+0.10}_{-0.08})$	9.2 @ 42*	-1.11 @ 32*	52
Delta 3	2017	C	183	128	32–40	-3.727	-7.43	$-1.11^{+0.03}_{-0.02}$	$-1.18^{+0.03}_{-0.04}$	$(-1.32^{+0.05}_{-0.08})$	9.5 @ 43*	-1.10 @ 32	52
Delta 4	2020	C	416	252	33–41	-3.152	-7.83	$-1.09^{+0.02}_{-0.02}$	$-1.14^{+0.02}_{-0.02}$	$(-1.24^{+0.03}_{-0.05})$	10.1 @ 45*	-1.09 @ 34	54
Factor 2	2011	C	110	87	32–40	-4.762	-7.06	$-1.16^{+0.04}_{-0.05}$	$-1.29^{+0.08}_{-0.07}$	$(-1.50^{+0.17}_{-0.17})$	8.6 @ 40	-1.12 @ 30*	54
Fusion	2019	C	165	111	35–43	-3.429	-8.07	$-1.15^{+0.03}_{-0.04}$	$-1.20^{+0.02}_{-0.03}$	$(-1.31^{+0.05}_{-0.07})$	9.5 @ 45*	-1.14 @ 34*	56
Lyght	2024	C	197	105	34–42	-3.346	-7.00	$-1.03^{+0.03}_{-0.03}$	$-1.09^{+0.02}_{-0.03}$	$(-1.21^{+0.05}_{-0.05})$	10.3 @ 43*	-1.02 @ 33*	56
Lynx	2018	C	117	100	33–41	-3.634	-7.98	$-1.16^{+0.04}_{-0.03}$	$-1.22^{+0.07}_{-0.04}$	$(-1.35^{+0.13}_{-0.10})$	9.3 @ 44*	-1.15 @ 33	52
Lynx 2	2023	C	124	91	32–40	-4.897	-6.31	$-1.10^{+0.05}_{-0.08}$	$-1.24^{+0.07}_{-0.05}$	$(-1.46^{+0.17}_{-0.11})$	9.0 @ 38	-1.04 @ 29*	52
Photon	2023	C	589	249	34–42	-3.439	-7.10	$-1.05^{+0.03}_{-0.02}$	$-1.11^{+0.01}_{-0.01}$	$(-1.24^{+0.03}_{-0.03})$	10.1 @ 43*	-1.04 @ 33*	56
Savage	2020	C	268	169	31–39	-4.499	-6.43	$-1.07^{+0.02}_{-0.02}$	$(-1.20^{+0.03}_{-0.03})$	$(-1.39^{+0.06}_{-0.06})$	9.3 @ 39*	-1.03 @ 30*	52
Sigma 7	2008	C	224	150	30–38	-5.184	-6.82	$-1.18^{+0.03}_{-0.03}$	$(-1.32^{+0.06}_{-0.07})$	$(-1.56^{+0.12}_{-0.15})$	8.4 @ 39*	-1.12 @ 29*	52
Sigma 8	2010	C	239	148	31–39	-4.420	-7.33	$-1.16^{+0.03}_{-0.03}$	$(-1.27^{+0.04}_{-0.05})$	$(-1.45^{+0.08}_{-0.10})$	8.8 @ 41*	-1.13 @ 31	50
Sigma 9	2014	C	304	219	31–39	-4.679	-6.90	$-1.14^{+0.03}_{-0.03}$	$(-1.26^{+0.05}_{-0.05})$	$(-1.47^{+0.08}_{-0.09})$	8.8 @ 40*	-1.10 @ 30*	50
Sigma 10	2017	C	284	177	32–40	-3.988	-6.97	$-1.08^{+0.03}_{-0.02}$	$-1.18^{+0.03}_{-0.05}$	$(-1.34^{+0.05}_{-0.12})$	9.5 @ 41*	-1.06 @ 32*	52
Sigma 11	2020	C	322	196	33–41	-3.435	-7.84	$-1.12^{+0.03}_{-0.02}$	$-1.18^{+0.02}_{-0.03}$	$(-1.30^{+0.04}_{-0.06})$	9.6 @ 44*	-1.12 @ 34	55
Sigma DLS	2024	C	148	93	33–41	-3.085	-8.44	$-1.15^{+0.03}_{-0.03}$	$-1.18^{+0.02}_{-0.04}$	$(-1.28^{+0.04}_{-0.08})$	9.8 @ 46*	-1.15 @ 35	52
Volt 3	2018	C	160	122	29–37	-5.174	-6.23	$-1.12^{+0.04}_{-0.03}$	$(-1.27^{+0.07}_{-0.08})$	$(-1.51^{+0.11}_{-0.15})$	8.8 @ 38*	-1.04 @ 29*	48
Volt 4	2022	C	367	204	31–39	-3.622	-7.26	$-1.08^{+0.01}_{-0.02}$	$(-1.15^{+0.02}_{-0.02})$	$(-1.29^{+0.04}_{-0.04})$	9.8 @ 43*	-1.07 @ 32	52
<i>EN B (48 types)</i>													
Apollo 2	2019	B	130	99	30–38	-4.334	-7.44	$-1.16^{+0.02}_{-0.03}$	$(-1.26^{+0.04}_{-0.07})$	$(-1.44^{+0.09}_{-0.13})$	8.8 @ 41*	-1.14 @ 31	47
Arak	2018	B	267	169	32–40	-3.652	-8.32	$-1.19^{+0.02}_{-0.02}$	$-1.25^{+0.03}_{-0.03}$	$(-1.38^{+0.06}_{-0.06})$	9.1 @ 44*	-1.19 @ 34	49
Arak Air	2021	B	339	181	31–39	-4.480	-7.40	$-1.17^{+0.02}_{-0.03}$	$(-1.28^{+0.03}_{-0.03})$	$(-1.47^{+0.06}_{-0.06})$	8.7 @ 41*	-1.15 @ 31	50
Base 2	2020	B	241	193	32–40	-4.217	-7.61	$-1.17^{+0.03}_{-0.03}$	$-1.26^{+0.04}_{-0.05}$	$(-1.43^{+0.08}_{-0.11})$	8.8 @ 42*	-1.15 @ 32*	50
Beat	2020	B	162	138	32–40	-4.453	-7.50	$-1.18^{+0.03}_{-0.03}$	$-1.29^{+0.06}_{-0.05}$	$(-1.47^{+0.10}_{-0.08})$	8.7 @ 41*	-1.16 @ 31*	49
Beat Light	2021	B	145	104	32–40	-3.959	-8.50	$-1.24^{+0.02}_{-0.04}$	$-1.31^{+0.03}_{-0.03}$	$(-1.45^{+0.07}_{-0.10})$	8.6 @ 44*	-1.23 @ 33	50
Chili 5	2021	B	148	116	32–40	-3.534	-8.49	$-1.20^{+0.04}_{-0.03}$	$-1.25^{+0.05}_{-0.04}$	$(-1.37^{+0.08}_{-0.10})$	9.1 @ 45*	-1.20 @ 34	50
Cumeo	2018	B	156	121	33–41	-3.249	-8.07	$-1.13^{+0.02}_{-0.02}$	$-1.17^{+0.02}_{-0.02}$	$(-1.28^{+0.05}_{-0.05})$	9.8 @ 45*	-1.13 @ 34	52

Table 3 continued

Type	Year	Class	Flights	Days	Window	a	b	$s(35)$	$s(40)$	$s(45)$	$(L/D)_{\max}$	$s_{\min}$	MS
Epsilon 6	2008	B	157	121	31–39	-3.956	-8.63	$-1.25^{+0.03}_{-0.03}$	$(-1.32^{+0.04}_{-0.06})$	$(-1.46^{+0.11}_{-0.14})$	8.6 @ 44*	-1.25 @ 33	52
Epsilon 7	2013	B	134	106	30–38	-3.439	-8.76	$-1.22^{+0.04}_{-0.05}$	$(-1.26^{+0.08}_{-0.12})$	$(-1.37^{+0.13}_{-0.21})$	9.1 @ 46*	-1.22 @ 35	52
Epsilon 8	2015	B	209	160	31–39	-4.018	-8.58	$-1.25^{+0.03}_{-0.03}$	$(-1.32^{+0.07}_{-0.06})$	$(-1.47^{+0.12}_{-0.12})$	8.5 @ 44*	-1.25 @ 33	48
Epsilon 9	2018	B	343	239	31–39	-4.799	-7.54	$-1.22^{+0.03}_{-0.04}$	$(-1.34^{+0.06}_{-0.04})$	$(-1.54^{+0.14}_{-0.09})$	8.3 @ 40*	-1.18 @ 31*	50
Epsilon DLS	2022	B	329	179	30–38	-3.847	-8.19	$-1.20^{+0.02}_{-0.02}$	$(-1.26^{+0.03}_{-0.03})$	$(-1.41^{+0.05}_{-0.05})$	8.9 @ 44*	-1.19 @ 33	48
Explorer	2017	B	254	184	32–40	-4.023	-7.35	$-1.12^{+0.01}_{-0.02}$	$-1.21^{+0.02}_{-0.02}$	$(-1.37^{+0.05}_{-0.05})$	9.2 @ 42*	-1.11 @ 32*	50
Explorer 2	2021	B	217	141	32–40	-3.432	-7.86	$-1.12^{+0.02}_{-0.02}$	$-1.18^{+0.02}_{-0.03}$	$(-1.30^{+0.05}_{-0.05})$	9.6 @ 44*	-1.12 @ 34	50
Geo 5	2016	B	104	84	31–39	-4.951	-7.00	$-1.18^{+0.03}_{-0.04}$	$(-1.31^{+0.07}_{-0.09})$	$(-1.53^{+0.14}_{-0.17})$	8.5 @ 39*	-1.13 @ 30*	50
Geo 6	2019	B	153	119	31–39	-2.786	-8.93	$-1.18^{+0.03}_{-0.03}$	$(-1.19^{+0.04}_{-0.04})$	$(-1.26^{+0.08}_{-0.07})$	10.0 @ 48*	-1.17 @ 37	50
Ikuma 2 P	2020	B	228	148	31–39	-3.458	-7.69	$-1.11^{+0.03}_{-0.03}$	$(-1.17^{+0.03}_{-0.06})$	$(-1.29^{+0.06}_{-0.09})$	9.7 @ 44*	-1.10 @ 33	50
Ikuma 3	2023	B	103	76	32–40	-3.034	-8.07	$-1.11^{+0.03}_{-0.02}$	$-1.14^{+0.02}_{-0.03}$	$(-1.24^{+0.04}_{-0.07})$	10.1 @ 46*	-1.11 @ 35	49
Ikuma 3 P	2023	B	288	145	31–39	-4.008	-7.04	$-1.09^{+0.03}_{-0.02}$	$(-1.18^{+0.03}_{-0.04})$	$(-1.35^{+0.06}_{-0.07})$	9.4 @ 41*	-1.07 @ 32	49
Ion 4	2016	B	188	133	33–41	-3.916	-8.14	$-1.20^{+0.03}_{-0.03}$	$-1.27^{+0.03}_{-0.03}$	$(-1.42^{+0.08}_{-0.06})$	8.8 @ 43*	-1.19 @ 33*	56
Ion 6 Light	2020	B	289	132	31–39	-3.885	-8.70	$-1.25^{+0.02}_{-0.02}$	$(-1.32^{+0.03}_{-0.04})$	$(-1.45^{+0.05}_{-0.07})$	8.6 @ 44*	-1.25 @ 34	48
Iota	2014	B	624	401	33–41	-3.198	-8.46	$-1.16^{+0.01}_{-0.01}$	$-1.20^{+0.02}_{-0.02}$	$(-1.30^{+0.03}_{-0.03})$	9.6 @ 46*	-1.16 @ 35	51
Iota 2	2017	B	412	240	32–40	-4.245	-7.00	$-1.11^{+0.03}_{-0.03}$	$-1.21^{+0.03}_{-0.03}$	$(-1.39^{+0.06}_{-0.05})$	9.2 @ 41*	-1.08 @ 31*	51
Iota DLS	2021	B	635	270	33–41	-3.612	-7.67	$-1.12^{+0.02}_{-0.02}$	$-1.19^{+0.02}_{-0.01}$	$(-1.32^{+0.03}_{-0.04})$	9.5 @ 44*	-1.11 @ 33	52
Maestro	2019	B	154	108	33–41	-3.669	-7.31	$-1.09^{+0.03}_{-0.04}$	$-1.16^{+0.05}_{-0.05}$	$(-1.30^{+0.09}_{-0.11})$	9.7 @ 43*	-1.08 @ 32*	51
Maestro X-Alps	2019	B	203	140	32–40	-4.406	-7.42	$-1.17^{+0.03}_{-0.03}$	$-1.27^{+0.03}_{-0.03}$	$(-1.45^{+0.07}_{-0.07})$	8.8 @ 41*	-1.14 @ 31*	52
Maestro 2 Light	2022	B	179	111	32–40	-4.259	-6.78	$-1.09^{+0.03}_{-0.05}$	$-1.19^{+0.04}_{-0.04}$	$(-1.37^{+0.12}_{-0.10})$	9.3 @ 40*	-1.06 @ 31*	52
Mentor	2008	B	283	191	33–41	-3.848	-8.59	$-1.24^{+0.05}_{-0.05}$	$-1.30^{+0.02}_{-0.03}$	$(-1.44^{+0.06}_{-0.08})$	8.7 @ 44*	-1.23 @ 33	54
Mentor 2	2010	B	259	153	32–40	-3.592	-8.02	$-1.16^{+0.02}_{-0.02}$	$-1.21^{+0.03}_{-0.03}$	$(-1.34^{+0.06}_{-0.06})$	9.3 @ 44*	-1.15 @ 33	54
Mentor 4	2014	B	153	118	34–42	-3.518	-7.71	$-1.12^{+0.03}_{-0.03}$	$-1.18^{+0.02}_{-0.03}$	$(-1.30^{+0.05}_{-0.06})$	9.6 @ 44*	-1.11 @ 33*	55
Mentor 5	2017	B	155	111	34–42	-3.611	-8.34	$-1.19^{+0.04}_{-0.03}$	$-1.25^{+0.04}_{-0.03}$	$(-1.37^{+0.08}_{-0.08})$	9.1 @ 44*	-1.19 @ 34*	54
Mentor 5 Light	2017	B	103	76	34–42	-3.688	-8.33	$-1.20^{+0.06}_{-0.05}$	$-1.25^{+0.04}_{-0.03}$	$(-1.39^{+0.08}_{-0.07})$	9.0 @ 44*	-1.19 @ 34*	52
Mentor 7 Light	2022	B	267	168	33–41	-4.413	-6.72	$-1.10^{+0.02}_{-0.03}$	$-1.21^{+0.04}_{-0.04}$	$(-1.40^{+0.10}_{-0.07})$	9.2 @ 40	-1.06 @ 30*	53
Mistral 4	2006	B	107	90	30–38	-4.915	-8.63	$-1.34^{+0.05}_{-0.02}$	$(-1.45^{+0.07}_{-0.14})$	$(-1.65^{+0.12}_{-0.32})$	7.7 @ 41*	-1.32 @ 32	50
Phantom	2016	B	229	186	31–39	-3.968	-8.11	$-1.20^{+0.02}_{-0.02}$	$(-1.27^{+0.03}_{-0.03})$	$(-1.42^{+0.06}_{-0.05})$	8.8 @ 43*	-1.19 @ 33	50
Rush 4	2014	B	162	117	32–40	-4.099	-7.15	$-1.11^{+0.02}_{-0.03}$	$-1.21^{+0.03}_{-0.04}$	$(-1.37^{+0.07}_{-0.08})$	9.2 @ 41*	-1.09 @ 31*	52
Rush 5	2018	B	153	123	32–40	-3.606	-7.20	$-1.07^{+0.05}_{-0.03}$	$-1.14^{+0.04}_{-0.03}$	$(-1.28^{+0.07}_{-0.07})$	9.8 @ 43*	-1.06 @ 32	51
Rush 6	2021	B	208	138	33–41	-3.249	-7.91	$-1.11^{+0.03}_{-0.02}$	$-1.16^{+0.03}_{-0.03}$	$(-1.27^{+0.05}_{-0.07})$	9.9 @ 45*	-1.11 @ 34	50
Soar	2020	B	328	235	32–40	-4.024	-7.59	$-1.15^{+0.03}_{-0.03}$	$-1.24^{+0.04}_{-0.03}$	$(-1.39^{+0.07}_{-0.07})$	9.1 @ 42*	-1.14 @ 32	51
Susi 3	2018	B	102	99	31–39	-4.463	-7.40	$-1.17^{+0.06}_{-0.09}$	$(-1.28^{+0.15}_{-0.15})$	$(-1.46^{+0.29}_{-0.26})$	8.7 @ 41*	-1.14 @ 31	52
Swift	2009	B	114	98	32–40	-4.991	-7.87	$-1.27^{+0.05}_{-0.06}$	$-1.39^{+0.04}_{-0.03}$	$(-1.60^{+0.09}_{-0.08})$	8.0 @ 40*	-1.23 @ 31*	52
Swift 4	2014	B	345	256	31–39	-4.482	-6.73	$-1.10^{+0.02}_{-0.02}$	$(-1.22^{+0.03}_{-0.04})$	$(-1.41^{+0.07}_{-0.07})$	9.1 @ 40*	-1.07 @ 30*	50
Swift 5	2018	B	495	260	32–40	-3.618	-7.30	$-1.08^{+0.02}_{-0.02}$	$-1.15^{+0.02}_{-0.02}$	$(-1.29^{+0.04}_{-0.05})$	9.7 @ 43*	-1.07 @ 33	51
Swift 6	2022	B	594	262	32–40	-3.905	-7.12	$-1.09^{+0.01}_{-0.01}$	$-1.18^{+0.02}_{-0.02}$	$(-1.35^{+0.04}_{-0.03})$	9.5 @ 42*	-1.07 @ 32*	54

Table 3 continued

[illegible]

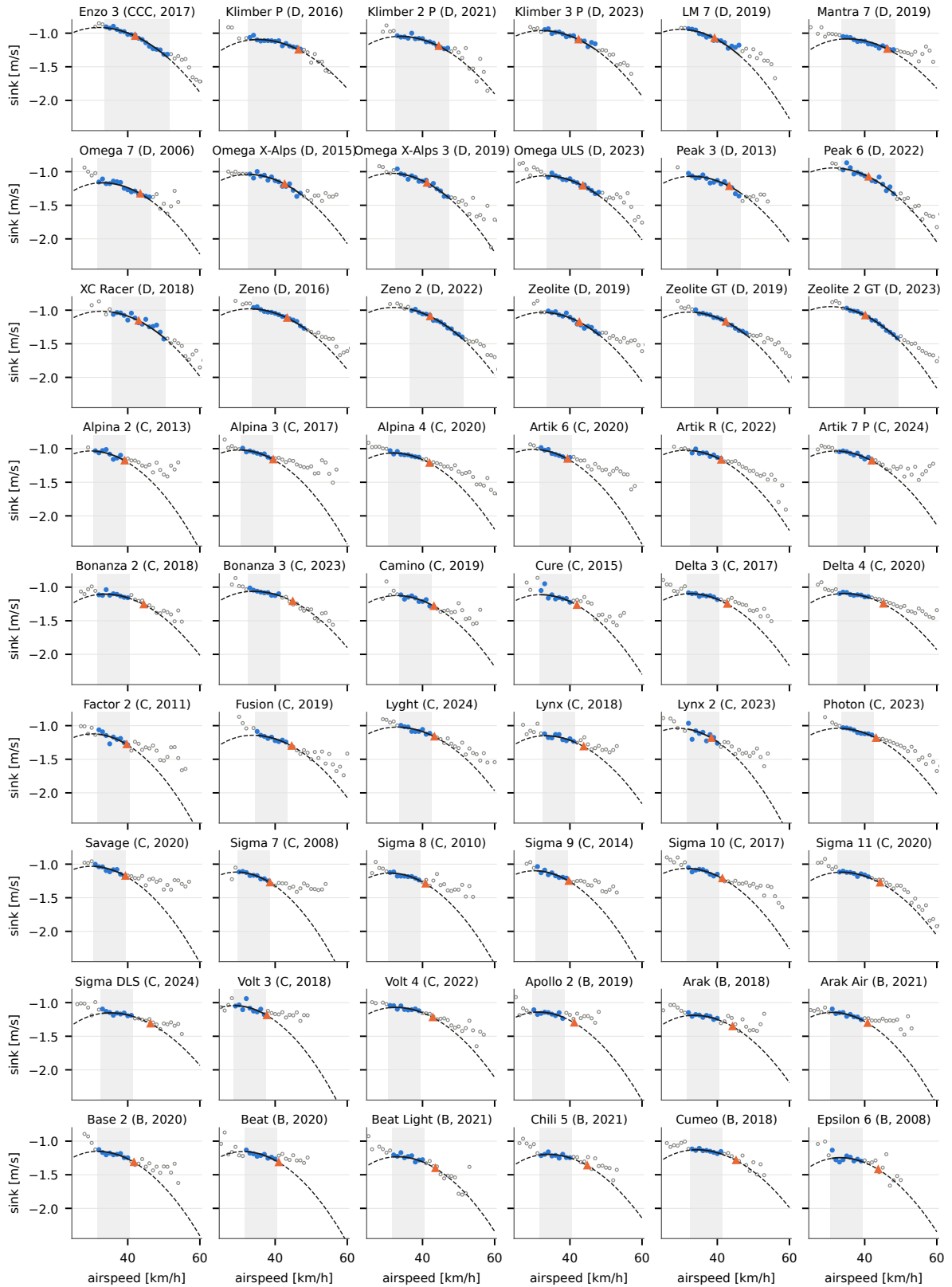


Figure 13: Polars of the glider types of Table 3, in the order of the table (part 1; part 2 in Figure 14). Points: most frequent value in 1 km h^{-1} bands at 2000 m after the subsidence correction; filled in the window of the type (shaded), open outside. Line: the physical curve fitted in the window, dashed outside. Triangle: best glide ratio of the curve. In brackets: class and year.

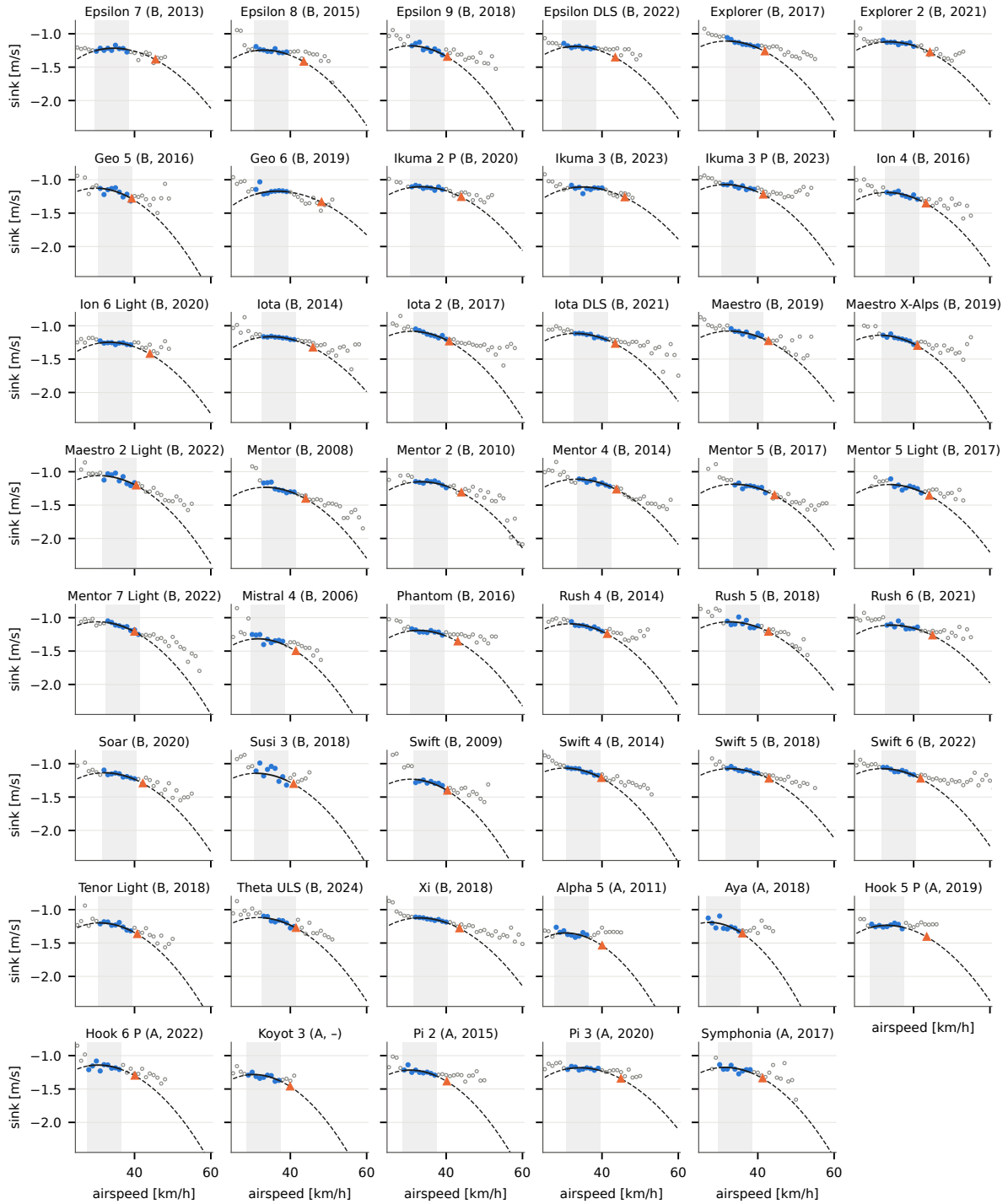


Figure 14: Polars of the glider types of Table 3, part 2; as in Figure 13.